



Advancements in talking head generation: a comprehensive review of techniques, metrics, and challenges

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Abstract

Talking head generation (THG) has emerged as a transformative technology in computer vision, synthesizing realistic human faces synchronized with audio, image, text, or video inputs. This paper systematically reviews THG methodologies and frameworks, categorizing approaches into 2D-based, 3D-based, Neural Radiance Fields (NeRF)-based, diffusion-based, and other techniques. We explore the most effective approaches in THG, emphasizing training techniques that improve realism, identity preservation, and motion accuracy. THG has vast potential applications, including creating digital avatars, dubbing videos, enhancing virtual assistants, and improving video calls. However, challenges include needing large models, handling extreme head movements, maintaining language synchronicity, and ensuring smooth visuals persist. This review provides a comprehensive overview of current progress, identifies ongoing challenges, and suggests future research directions, including developing simpler and more adaptable models, enabling real-time processing on smaller devices, and creating ethical guidelines. By summarizing existing research and highlighting ongoing challenges, this overview offers valuable insights for anyone interested in future of talking head technology. For the complete survey, code, and curated resource list, visit our GitHub repository: <https://github.com/VineetKumarRakesh/thg>.

Keywords Talking head generation · Deep learning · Systematic review · Evaluation metric

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1 Introduction

Deep learning (DL) and artificial neural network (ANN) have altered computer vision, enabling breakthroughs in the area of talking head generation (THG) and synthesis of realistic human faces for speech articulation [1]. THG is a research subject that seeks to generate realistic video pictures of human faces that speak in synchronization with arbitrary audio, image, text, and others. as input. Today, this technology finds uses in digital avatars, video dubbing in movies, online schooling, and video conferencing [2].

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Early approaches concentrated on 2D-based algorithms applying deep generative models, but they failed to capture 3D structural information, which resulted in less realistic facial dynamics and perspective modifications [2]. Recent breakthroughs in 3D scene representation approaches, notably Neural Radiance Fields (NeRF) [3], have boosted realism, allowed free-view control for movies, and improved picture quality. Innovative frameworks like Wav2NeRF [2] have emerged to tackle issues such as correctly morphing lip movements in rhythm with audio and managing high-frequency details [2]. The subject has developed from early rule-based approaches to sophisticated deep learning methods employing Generative Adversarial Networks (GANs) and attention processes [1].

The synthesis of talking heads encompasses portrait production, driving mechanisms, and editing techniques, enabling the adjustment of attributes such as emotion, head posture, and eye blinking [4]. This study thoroughly examines the existing methodology and approaches for creating THG and assessing algorithms, datasets, and evaluation matrices. It arranges processes into separate groupings, stressing their contributions and drawbacks. Focusing on technical efficiency and perceptual realism is crucial for real-time interaction and high visual quality applications—the research also provides a comparative review of publicly accessible technologies for THG.

Portrait production has improved dramatically by utilizing generative techniques, including unconditional and conditional methods. Unconditional techniques provide random visuals without prior labels or data inputs, while conditional methods offer controlled outputs based on descriptive inputs [4]. However, making high-quality, realistic portraits remains tricky due to intricate geometry and appearance. Talking head synthesis, a driving mechanism for animating facial characteristics, has shown remarkable performance utilizing audio-driven and video-driven techniques. Advanced deep learning algorithms have boosted real-time rendering capabilities, yet ongoing constraints like temporal inconsistency and training data necessitate continued effort. Editing in talking head synthesis enables control over avatar modification, although decoupling is challenging due to the connected nature of attributes.

Different sorts of deep learning architecture are utilized to implement any model in THG. Deep learning architectures, such as convolutional neural network (CNN) [5] and recurrent neural network (RNN) [6], have proved crucial in AI breakthroughs [7]. CNNs, first created for image processing [7], have proved their capacity to tackle tough visual problems with better accuracy than earlier approaches [8]. RNNs, on the other hand, are created to handle sequence data and have applications in voice recognition and language modeling [9]. Including LSTM units has increased the efficacy of these models by eliminating the vanishing gradient issue [7].

The universal approximation property of neural networks, initially established by Hornik et al. [10], states that a neural network with a single hidden layer may approximate any continuous function on a compact subset of $\mathbb{R}^n$ to any desired degree of accuracy. This theorem explains why even primitive systems may convey intricate relationships in data. Neural networks are often divided into shallow and deep learning models, with shallow networks having one or two hidden layers, limited in their ability to learn complex features. In contrast, deep neural networks with multiple hidden layers can perform hierarchical feature extraction and more sophisticated data representations [7].

Talking head video creation includes synthesizing lip motion sequences matching a driving source, such as voice or text. This technique also evaluates facial features like emotions and head movements. Early techniques employed cross-modal retrieval [11] and HMM-based algorithms [12], but they had high requirements. Recent breakthroughs in deep learning technology have permitted talking head video generation approaches, utilizing 2D and 3D-based processes. A considerable amount of data must be applied for training, testing, and validation to train a model, build a talking head employing the above-mentioned approaches, or even employ pre-existing models properly. Accuracy rises with the quantity of data, but only up to a limit; beyond that, fresh data may bring declining returns in performance. To ensure that the model does not start memorizing the input, rather, the model learns, backpropagation, invented by Linnainmaa [13], is a method used to change link weights in neural networks to reduce learning errors. It was popularized in the 1980s and was later developed to correct mistakes during learning. Based on automated differentiation, this technique assures that the model learns without memory retention. It distributes error amounts across connections and determines the gradient of the cost function. Weight updates may be done using techniques such as stochastic gradient descent [14], extreme learning machines [15], no-prop networks [16], weightless networks, and non-connectionist neural networks.

Publicly available datasets have allowed for important improvements in talking head synthesis. The model we have studied indicates that VoxCeleb [17], VoxCeleb2 [18], and TH-1KH [19] are among the most commonly used datasets for video-driven tasks, while CREMA-D [20], LRW [21], and MEAD [22] have become prominent datasets for audio-driven tasks, according to our several recent surveys on talking head generation. Large-size, high-resolution datasets like FFHQ [23], HDTF [24], and CelebV-HQ [25] evolved as a result of the growing necessity for higher image quality brought about by the rise of application scenarios. In-depth details about each dataset, such as picture size, modality, and subject perspective, have been acquired; these factors have not been fully investigated in past evaluations of talking head synthesis.

While training a model, a critical component in training models for optimal accuracy is the design and implementation of a robust loss function, which assesses the difference between anticipated outputs and actual ground truth. A loss function is used in training models to improve parameters by examining the difference between anticipated and expected outcomes [26]. Although convexity assures that any local minimum is also a global minimum, which is theoretically ideal, many effective deep learning loss functions are non-convex. However, they may be efficiently improved utilizing modern approaches [26]. An efficient loss function is created to be resilient to outliers and differentiable to allow smooth gradient-based optimization. Convex loss functions are popular since they may be optimized using gradient-based approaches. Differentiability is vital for allowing gradient-based optimization. Robustness indicates that loss functions can support outliers without being influenced by a small number of extreme values. Smoothness provides a steady gradient without sudden transitions or spikes. Sparsity encourages sparse output, suited for high-dimensional data and tiny features. In non-convex conditions, a monotonic drop in the loss function does not assure convergence to the optimal solution, even if it would demonstrate consistent optimization progress.

As per Chen et al. [27], the performance of talking head synthesis models is significantly examined based on four fundamental criteria which are identity preservation, visual quality, lip synchronization, and natural motion. However, models usually focus on increases in a single component and qualitative and quantitative assessments are commonly applied. Quantitative measures offer a more objective analysis, but qualitative judgments rely on direct observation and could be subjective. This section includes some well-known quantitative criteria for a complete examination of talking head synthesis models.

Recent advances in perceptual evaluation have introduced THQA [28], a specialized metric designed to assess human-perceived visual quality in talking head generation (THG). Unlike traditional metrics such as PSNR, SSIM, and LPIPS, which primarily focus on pixel-level fidelity, THQA [28] evaluates multi-dimensional perceptual realism, integrating structural coherence, expression consistency, and naturalness of lip–audio synchronization. Moreover, SyncNet-based accuracy remains essential for evaluating audio-visual alignment, while FID captures generative diversity, and LPIPS measures perceptual similarity between reference and synthesized outputs. In multilingual contexts, evaluation becomes more challenging due to phoneme diversity, prosodic variations, and cultural lip articulation patterns. Datasets such as LRS3-TED [29] and Face-Dubbing++ [30] have enabled multilingual word error rate (mWER) quantification, providing insights into cross-lingual intelligibility and prosody preservation. However, existing metrics inade-

quately capture accent-driven mismatches, speaker-specific expression gaps, and temporal prosody drift across languages. Integrating THQA [28] with mWER benchmarks creates a comprehensive evaluation protocol, particularly valuable for multilingual dubbing and cross-lingual avatar synthesis. Future benchmarking efforts should combine objective quality assessments with subjective human preference studies to better quantify user-perceived realism across diverse linguistic and cultural settings.

Additionally, the multilingual component of talking head synthesis is another problem. The paucity of annotated, high-quality datasets across numerous languages, with diverse phonetic patterns, lip movements, and cultural intricacies, inhibits the general deployment of these systems [31]. To tackle this challenge, researchers should foster multilingual datasets and employ sophisticated methodologies like self-supervised and transfer learning. Complementary options include approaches like Long Short-Term Memory (LSTM) [32] networks to acquire mouth landmarks from audio, the Wav2Lip model [33] to automate lip-syncing, and a strategy for reenactment that stresses visual signals for managing different voices and modifying face gestures depending on audio-derived speech styles [34].

The methodical technique clarifies the present situation and brings intriguing prospects for further inquiry. This article seeks thorough insights, including for researchers in the THG research topic. Despite the great benefits of THG, it tackles several issues, such as its dependence on pre-trained models, the management of extreme postures, and the necessity for high-quality datasets and sophisticated algorithms [31]. These limits may hamper innovation and adaptation for varied use cases.

A critical refinement in analyzing pre-trained model dependency is the clear separation between architectural reliance and dataset bias. Models such as MetaPortrait [35] and MegaPortraits [36] demonstrate strong architectural reliance by leveraging pre-trained generative priors—for example, StyleGAN2-based texture encoders [37] or ImageNet [38]/VGG-Face [39] initialized identity modules to enhance sample efficiency and accelerate convergence. While these priors improve identity preservation and visual fidelity, they also constrain architectural diversity, limiting explorations of novel generative formulations. Conversely, dataset bias stems from imbalanced linguistic and cultural distributions within widely adopted benchmarks. Popular datasets such as VoxCeleb [17] and LRS3-TED [29] are predominantly English-centric, which restricts cross-lingual generalization, impacting phoneme coverage, prosody modeling, and multilingual dubbing quality. Addressing these limitations requires expanding curated datasets toward broader phonetic inventories and accent diversity, as seen in emerging efforts like Face-Dubbing++

[30], which aim to improve cross-lingual intelligibility and support equitable benchmarking of THG systems.

Future research could consider training individual components or modules on large-scale, diverse datasets to develop more modular and adaptable frameworks. Hybrid architectures that blend pre-trained models with task-specific layers or modules may give both pre-trained knowledge and targeted flexibility advantages. To solve problems such as large-angle poses, researchers aim to enhance datasets and apply multi-view training methodologies to capture a larger range of face orientations [31]. Maintaining temporal consistency is critical for providing smooth visual outputs, and tackling this involves high-quality datasets and innovative approaches [31].

The field of THG is experiencing consistent growth each day. Figure 1 illustrates the annual number of reported works in this area based on data collected from Google Scholar, including publications from academic publishers, professional societies, online repositories, universities, and other relevant websites under the keyword “Talking Head Generation” and its associated terms. For 2025, the publication count covers the period from January through April.

The united technical lineage of paradigms from 2017 to 2025 emphasizes the constant expansion of approaches. Initial strategies rely on 2D techniques, followed by the emergence of 3D technology and the creation of NeRF-driven pipelines. Subsequent improvements featured diffusion-based approaches and real-time rendering technologies, which substantially enhanced visual realism and efficiency. The diagram incorporates recent breakthroughs in light-field representations and their implementation into hybrid unified frameworks. The merging of NeRF-based and diffusion-driven paradigms reveals cross-fertilization tendencies, as modern models gradually amalgamate complementing traits. Additionally, two key issues are highlighted: architectural reliance, deriving from dependence on pre-trained generative priors, and dataset bias, resulting from skewed or limited training data distributions. Figure 2 illustrates the unified technical lineage THG paradigms from 2017 to 2025, stressing the continual progress of methodologies.

A comprehensive framework has been developed to classify highly cited works in THG according to input modality, model architecture and training paradigm.

- An in-depth examination of three fundamental components—innovation and advancement—portrait synthesis, driving mechanisms, and editing operations—has illuminated the inherent trade-offs among identity preservation, visual realism and user control.
- This analysis identifies several important research gaps, notably in multilingual synthesis, pose diversity, temporal coherence and the potential over-reliance on large pre-trained networks. To address these shortcomings, the

study encourages the exploration of modular architectures and the adoption of more varied training strategies.

Although several recent overviews Gowda et al. [1] and Nguyen-Le et al. (2024) [40] provide broad surveys of 2D/3D pipelines up to 2024, but offer limited discussion of emerging neural radiance field and diffusion-based approaches. Dedicated NeRF reviews [3] and general 3D generation summaries [41] tend to treat audio-driven lip-sync and modular design only peripherally. At the same time, security-focused deepfake detection works [31, 40] do not fully trace the evolution of underlying generative methodologies. Similarly, existing evaluations of perceptual metrics [42] and foundational THG principles [27] predate recent advances in real-time rendering and temporal consistency. By contrast, the present survey spans innovations from 2017 through 2025, offering detailed architectural and loss function comparisons across ten distinct THG paradigms and explicitly linking each method to a dozen practical applications ranging from video conferencing and remote education to visual effects. It is also the first to unify 3D Gaussian splatting and latent consistency models within a single analytical framework, and it proposes a set of ethical safeguards—such as robust watermarking protocols and the curation of multilingual datasets—to support responsible deployment of THG technologies.

This article outlines development main ideas of talking head, research scope, ethical difficulties, and methods. It describes the architecture of a proposed system and discusses training and inference processes, the dataset, loss function, and evaluation metrics. The overall structure of this review paper on **talking head generation (THG)** is organized into several well-defined sections for clarity and comprehensive coverage. It begins with the **Introduction** (Sect. 1), which outlines the motivation, background, and scope of the study. This is followed by the **Comparative Analysis** (Sect. 2), where existing approaches and frameworks are critically evaluated. Next, the paper presents a detailed **Taxonomy of Evaluation Metrics** (Sect. 3), highlighting both quantitative and qualitative measures used to assess THG systems. Subsequently, the **Challenges and Future Directions** (Sect. 4) section discusses open research problems and emerging trends in the field. The paper concludes with the **Conclusion** (Sect. 5), which summarizes the key findings and insights, while also outlining potential opportunities for

2 Comparative analysis of THG approaches

Walter Pitts and Warren McCulloch introduced the foundational concepts of deep learning (DL) and artificial neural networks (ANNs) in 1943 [43], who showed the capacity

Fig. 1 Number of papers published per year with the keyword “Talking Head Generation” from 2015 to 2025*. *Data for 2025 includes publications up to August, 2025.

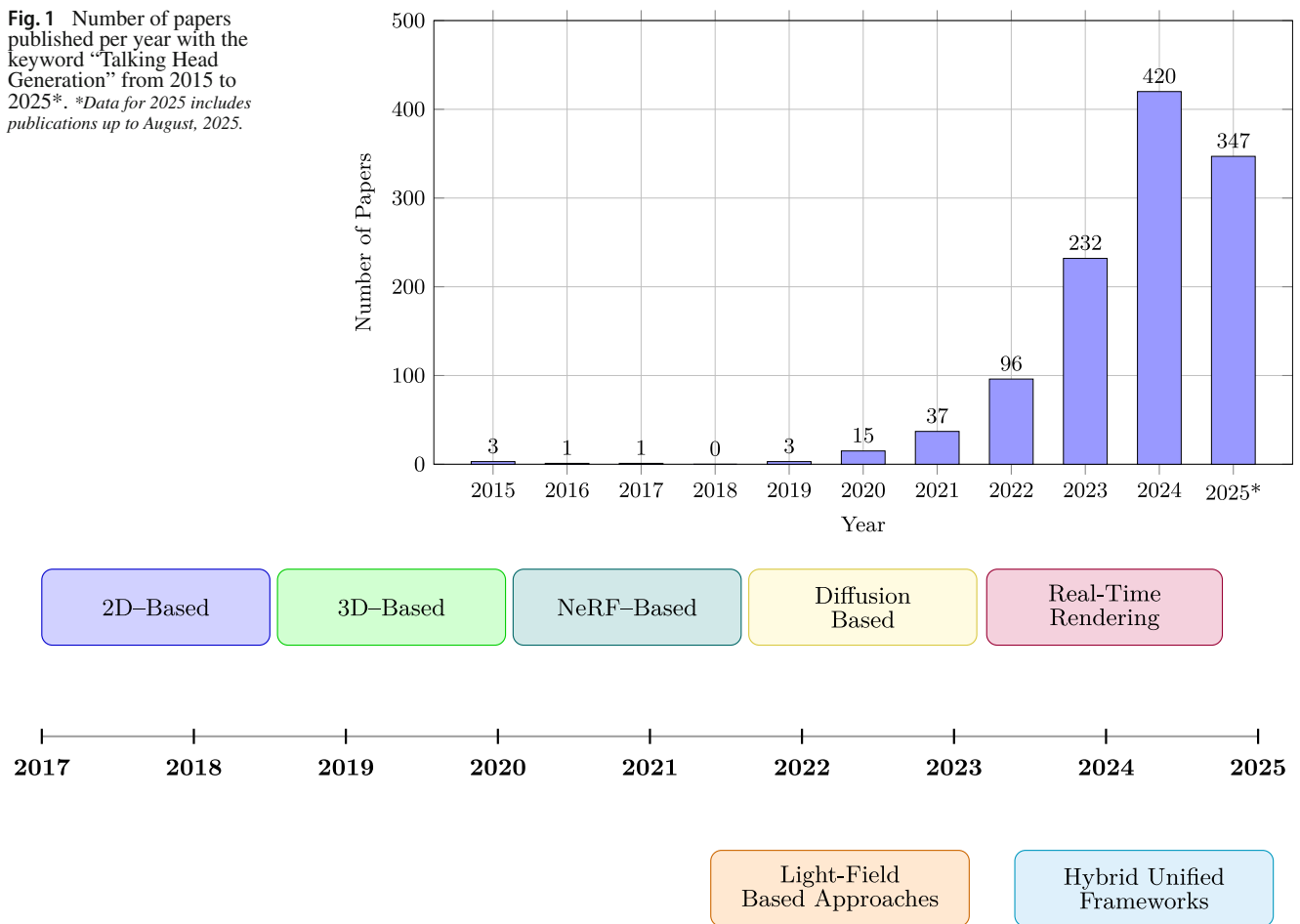


Fig. 2 Technical lineage of THG paradigms (2017–2025)

of theoretical artificial neuron networks to complete basic logical tasks. However, recent developments in processing power, notably the introduction of high-performance graphics processing units (GPUs), have substantially facilitated the deployment of complicated deep learning models, such as those applied in THG [44]. A examination of hundreds of articles on talking head generation (THG) revealed approximately 100 approaches for making talking heads from trustworthy academic sources. Picture-based, audio-based, text-based, video-based, 2D-based, 3D-based, distortion-based, Neural Radiance Fields (NeRF)-based, and 3D animation were the ten key areas in which their performance was evaluated. By analyzing their designs, datasets, benefits, drawbacks, and applications, the study separated these methods into ten paradigms. Figure 2 provides a conceptual overview of the technical lineage of talking head generation, while Fig. 3 complements it by summarizing the cross-paradigm innovation pathways, highlighting cross-fertilization trends and the progression toward real-time, multilingual, and fully controllable avatar synthesis.

The evolution of THG paradigms demonstrates a clear innovation trajectory from 2D-based geometric keypoint modeling toward 3D-aware generative pipelines and beyond. Early works such as FOMM [45] relied on local affine transformations and self-learned motion keypoints, achieving fast adaptation but lacking depth-awareness and pose generalization. These limitations motivated the emergence of 3D Morphable Models and volumetric motion modeling, exemplified by frameworks like Talk3D [46] and PV3D [47], which reconstruct explicit 3D structures for view-consistent head synthesis. Subsequent breakthroughs integrated Neural Radiance Fields (NeRFs) for free-view rendering, enabling geometry-preserving pipelines such as HiDe-NeRF [48] and DFA-NeRF [49], which improved photorealism but incurred heavy rendering costs. Recent paradigms fuse NeRF representations with diffusion-based generative priors to balance fine-grained realism, identity consistency, and temporal smoothness, achieving state-of-the-art results in PV3D [47] and MuseTalk [50]. Crucially, the paper distinguish architectural reliance from dataset bias such as models like MetaPortrait [35] inherit priors from StyleGAN, constrain-

ing generalization, whereas datasets such as VoxCeleb [17] introduce an English-centric skew that impacts multilingual naturalness.

2.1 Image-based THG

Image-based talking head generation (THG) animates static portraits by transferring motion information from a driving video. It synthesizes realistic head movements and facial expressions using self-learned keypoints, local affine transformations, an occlusion-aware generator, and an extended equivariance loss. Figure 4 shows a generalized architecture combining GAN-based synthesis with landmark-driven motion transfer.

Recent advancements in image-based THG leverage generative adversarial networks (GANs) to achieve high-fidelity facial animations. Lightweight GANs improve visual quality while reducing computational costs [51], and edge-aware CNNs enhance landmark detection and identity preservation [52]. DiscoFaceGAN [53] introduces a disentangled framework using 3D Morphable Models (3DMMs) and contrastive learning, enabling fine-grained control over identity, pose, and expression.

Table 1 compares key image-based THG approaches, including architectures, datasets, highlights, and limitations.

As shown in Table 1, SMA [54] uses VQGANs to jointly learn motion and appearance representations, producing smooth, natural motion flows. TS-Net [56] adopts a dual-branch design for identity retention and robust occlusion handling, while DaGAN [60] integrates self-supervised geometry learning for improved depth consistency but struggles with extreme poses and occlusions. SAFA [61] combines 3DMM-driven structure awareness to achieve higher perceptual realism, whereas Face2Face [11] supports real-time expression transfer for interactive applications. CrossID-GAN [62] improves multi-identity face reenactment using accurate landmark mapping but depends heavily on precise detection. MarioNETte [67] employs U-Net architectures with attention-based identity preservation, though performance degrades under large pose variations. FOMM [45] and AAO [63] propose flexible “Monkey-Net” architectures to decouple motion and appearance, enabling generalizable image animation beyond facial inputs. ReenactGAN [68] achieves photorealistic reenactment via boundary space transfer but is computationally intensive. DiscoFaceGAN [53] further enhances synthesis by leveraging explicit parametric control over facial dynamics but remains sensitive to lighting and 3DMM fitting errors. Integrating DiscoFaceGAN with NeRF-based free-view synthesis could improve multi-view consistency and high-resolution fidelity. X2Face [69] enables self-supervised pose and expression manipulation, while GATH [70] uses continuous action unit (AU) coefficients for expression synthesis, raising important

ethical considerations. Recent advancements extend image-based THG into real-world applications. Tencent’s HunyuanPortrait [71] and ByteDance’s X-Portrait [72] focus on ultra-realistic portrait animation. AVFR-GAN [73] integrates audio-visual cues for synchronized reenactment, while Samsung AI’s MegaPortraits [36] produces megapixel-resolution avatars. Emerging techniques, including LivePortrait [74], MCNET [75], TPSM [76], StyleHEAT [77], AniFaceGAN [78], and LIA [79], emphasize real-time animation, high-fidelity synthesis, and editable visual outputs, pushing image-based THG toward highly interactive applications. Additionally, as illustrated in Fig. 3, early image-based methods pioneered local affine transformations and keypoint-based motion transfer, enabling identity-agnostic portrait animation. This innovation inspired frameworks that integrated 3D Morphable Models and disentangled facial control, followed by geometry-guided constraints that improved perceptual realism. These advancements laid the groundwork for volumetric pipelines and eventually hybrid paradigms, where 2D motion cues are fused with 3D motion modeling for view-consistent realism.

2.2 Audio-based THG

Audio-driven THG generates synchronized lip movements and facial expressions directly from speech, leveraging cross-modal alignment and acoustic features. Figure 5 illustrates a generalized hybrid framework integrating GAN, NeRF, and diffusion-based architectures with landmark-guided motion estimation to produce temporally consistent, photorealistic talking heads.

This unified framework (Fig. 5) extracts phonetic and prosodic features from audio using an Embedder, which guides a Generator to synthesize synchronized frames while a Discriminator ensures realism through adversarial learning. Training involves minimizing content loss, aligning synthesized lip motions and expressions with ground truth, and match loss, promoting indistinguishability between generated and real frames. Table 2 compares state-of-the-art audio-based THG techniques, detailing architectures, datasets, advantages, and limitations.

As shown in Table 2, OmniHuman-1 [80] employs a diffusion transformer framework with motion conditioning for highly realistic results, while Flow-guided [84] enhances motion smoothness but faces style consistency issues. EMO [85] uses ReferenceNet to improve audio-facial alignment, though its performance drops with complex expressions. Talk3D [46] integrates 3D priors with audio-guided U-Nets for view-consistent synthesis but suffers from preparation complexity and generalization gaps. Models like MODA [94] introduce dual-attention fusion for enhanced synchronization but incur high computation costs. GC-AVT [98] enables fine-grained control of lip, pose, and expression

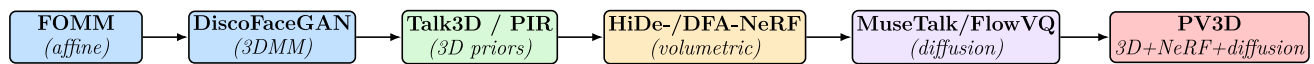


Fig. 3 An overview of the innovation pathways in talking head generation, illustrating the progressive evolution of techniques toward highly realistic and controllable synthesis frameworks

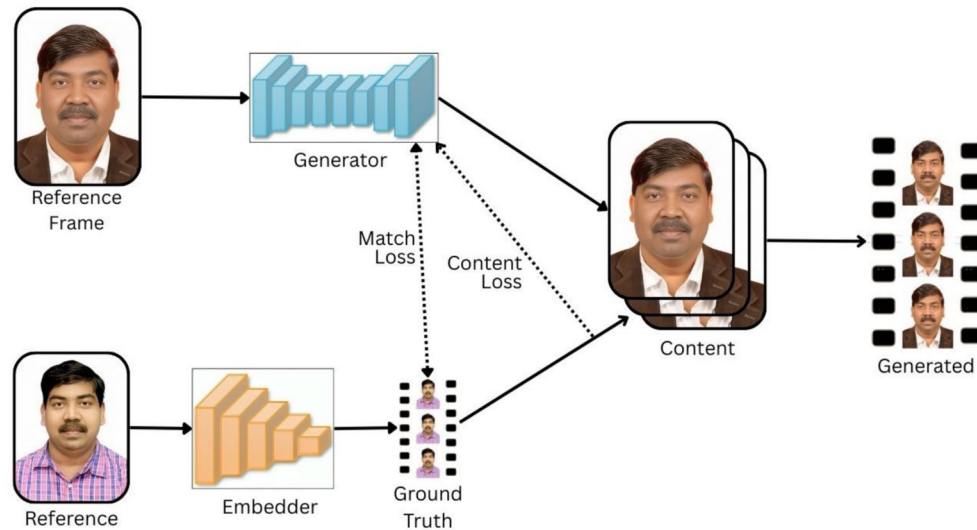


Fig. 4 Generalized approach for image based

Table 1 Comparison of image-based approaches

Method	Arch.	Dataset	Highlights	Limitations	N-shot
SMA [54] (2025)	VQGAN	VoxCeleb1 [55], CelebV-HQ [25]	Joint motion + appearance code-books for smooth motion flow	Keypoint motion estimation needed to reduce appearance leakage	One
TS-Net [56] (2022)	GAN	FaceForensics	Dual-branch with warp-free and GRID [57] transformation	Inconsistent motion, lacks high-freq detail	One
NeuralHead-FF [58] (2022)	FLAME	Custom Dataset	Monocular RGB Videos' Neural Head Avatars	Restrictions on the specified geometry	One
PhotoHelper [59] (2022)	ResNet	PhotoHelper	Novel portrait photographing guidance system that leverages deep feature retrieval and fusion	Focuses primarily on composition and aesthetics, potentially overlooking other important factors	One
DaGAN [60] (2022)	GAN	VoxCeleb1 [55], CelebV	Self-supervised geometry for synthetic faces	Quality degrades with poor input; costly	One
SAFA [61] (2021)	FAN	Voxceleb1	3DMM-driven structure-aware animation	Sensitive to occlusion, over-fitting	One
Face2Face [11] (2020)	CNN	Face2Face	Real-time expression transfer from monocular view	Lambertian assumptions + latency issues	One

Table 1 continued

Method	Arch.	Dataset	Highlights	Limitations	N-shot
CrossID-GAN [62] (2020)	GAN	300VW	Multi-ID reenactment with landmark mapping	Needs real-time and accurate landmarks	One
FOMM [45] (2020)	Monkey-Net	Multiple	Decouples appearance/motion; handles occlusion	Requires object-specific tuning + data	One
DiscoFaceGAN [53] (2020)	3DMM	FFHQ [23]	Imitative + contrastive learning with 3D priors	Sensitive to lighting/pose; lacks gaze control	One
AAO [63] (2019)	Monkey-Net	Multiple	Dense motion + keypoint prediction	Fails with poor keypoints	One
MeshG [64] (2020)	GCN	FaceWarehouse [65], AFLW [66]	Using GCN for Mesh-Guided Reenactment	Handling large pose variations, expression diversity, and identity preservation	One
MarioNETte [67] (2019)	U-Net	VoxCeleb1 [55], CelebV	Identity via image attention + landmark transform	Pose and detection errors limit scale	One
ReenactGAN [68] (2018)	GAN	WFLW, CelebV	Reenactment via boundary space transfer	Slow, data hungry	One
X2Face [69] (2018)	U-Net	VGG-Face, Vox-Celeb [17]	Self-supervised pose/exp control	Poor generalization to large poses	One
GATH [70] (2018)	CNN	CACD, GTAV	AU-based expression synthesis	Ethical concerns + data load	One

but struggles with background consistency. Wav2Lip-HR [96] achieves high-quality lip synchronization yet demands significant computational resources. PIR [97] combines parametric modeling with implicit neural fields for photorealism, while Audio2Head [100] predicts head pose and expressions from speech but faces identity mismatch challenges. DAVS [101] improves audio-driven realism for one-shot learning but risks overfitting when training data is scarce. VisemeNet [103] generates viseme-based speech motion curves for better lip-sync stability, though emotion diversity remains limited. FEP [104] combines hidden Markov models and DNNs to map speech to lip motions but requires large datasets and struggles with nuanced emotions. Recent models extend capabilities further. ACTalker [105] uses multimodal control signals—including pose, emotion, and speech—for high-fidelity diffusion-driven synthesis. AniPortrait [106] transforms static portraits into expressive, speech-driven animations. EDTalk [107] improves emotional disentanglement, while EchoMimic [108] mimics vocal subtleties to generate lifelike expressions. MuseTalk [50] achieves real-time, high-fidelity lip-syncing even for unseen speakers, while

FlowVQTalker [109] preserves emotional richness and identity consistency using flow-based quantization. ReSyncer [110] enhances lip-sync adaptability across arbitrary models, and Real3DPortrait [111] reconstructs animatable 3D avatars from single images guided by speech. ManiTalk [112] introduces explicit facial control over micro-expressions like eyebrow and pupil movement while integrating PIR [97] for geometry preservation and FEP [104] for improved temporal lip accuracy. However, PIR requires large-scale 3D datasets, and FEP struggles with emotional variability. THQA [28] proposes a perceptual metric to evaluate synthesized outputs, enabling consistent quality assessment across audio-driven THG systems. Collectively, audio-based THG has advanced toward highly expressive, real-time, and cross-lingual synthesis, but challenges remain in scaling to unseen speakers, handling emotional diversity, and balancing fidelity with computational efficiency. Moreover, as depicted in Fig. 3, audio-driven THG evolved from phoneme-aligned convolutional mappings into models that incorporate explicit 3D priors for enhanced lip synchronization. Recent advances combine phoneme-conditioned diffusion with 3D-

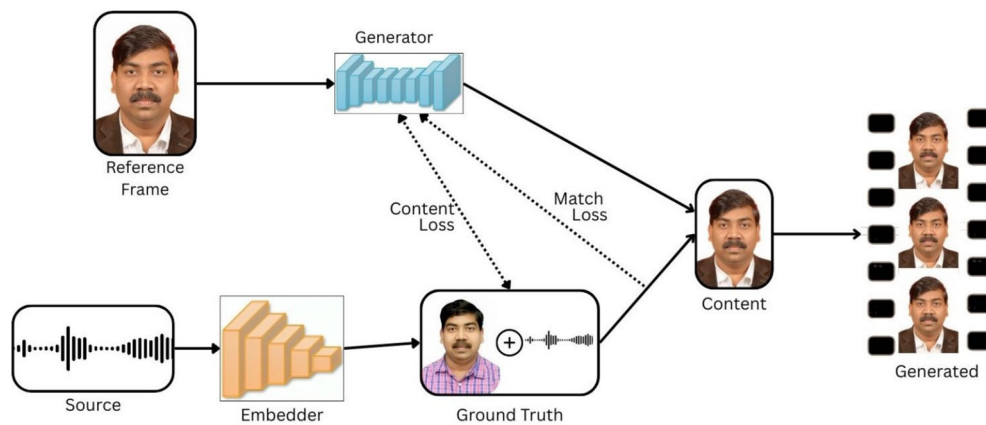


Fig. 5 Generalized approach for audio-based

Table 2 Comparison of audio-based approaches

Method	Arch.	Dataset	Highlights	Limitations	N-shot
OmniHuman-1 [80] (2025)	3DVAE	CelebV-HQ [25], RAVDESS [81]	Diffusion Transformer with motion conditioning for realism	Scalability, data dependency, and ethical concerns	One
X-Actor [82] (2025)	GAN	HDTF [24], CelebV-HQ [25]	Long-Range Emotional and Expressive Portrait Acting from Audio	Does not represent full-body motions or gestures	One
Sonic [83] (2025)	GAN	HDTF [24], CelebV-HQ [25]	Global Audio Perception in Portrait Animation	Preserving fine-grained lip-sync accuracy	One
Flow-guided [84] (2025)	3DMM	HDTF [24]	AV flow-guided with motion realism	Coherence, cropping, and style gaps	One
EMO [85] (2024)	RefNet	CREMA	Audio-facial correlation enhances realism	Expression complexity, quality, dependency	One
V-Express [86] (2024)	U-Net	HDTF [24], VFHQ [87]	Conditional dropout for progressive training in portrait video generation	The latest version works better with English but not as well with other languages	One
Talk3D [46] (2024)	GAN	AD-NeRF [88], Obama Weekly	Audio-driven 3D prior and attention U-Net	Prep complexity, artifacts, poor generalization	Multi
Audio2AB [89] (2024)	GAN	BEAT [90]	Create synchronized full-body animations and facial blendshape parameters	real-time deployment may be computationally	Multi
VASA-1 [91] (2024)	HFDG	VoxCeleb2 [18]	Real-time, life-like talking faces driven by audio	Processing only torso regions, causing artifacts	Multi

Table 2 continued

Method	Arch.	Dataset	Highlights	Limitations	N-shot
JOYVASA [92] (2024)	GAN	CelebV-HQ [25], HDTF [24]	Diffusion-based audio-driven facial dynamics in portrait and animal image animation	Generated upper bound is determined by the first-stage model's performance	One
Hallo [93] (2024)	ReferenceNet	HDTF [24], Custom Dataset	Animation of Portrait Images using Hierarchical Synthesis	Synchronization, coherence, efficiency, and diversity control for output	One
MODA [94] (2023)	LSGAN	HDTF [24], LSP [95]	Dual-attention and facial composer	Expression variance, compute cost	One
Wav2Lip-HR [96] (2023)	LSTM	LRS2	Maintain accurate lip synchronization	Computational cost for outputs may be higher	Multi
PIR [97] (2023)	3DMM	HDTF [24]	Combines parametric and implicit models for quality reenactment	High complexity, data dependence, ethical concerns	One
GC-AVT [98] (2022)	GAN	VoxCeleb2 [18], MEAD [22]	Granular control over lip, pose, and expression	Low-res output and poor backgrounds	Multi
FACIAL [99] (2021)	GAN	Custom Dataset	Integrating Implicit Attribute Learning with Dynamic Talking Face	Struggles with maintaining consistent facial attributes	Multi
Audio2Head [100] (2021)	RNN	LRW, GRID [57], VoxCeleb [17]	Predicts pose and motion from audio/image	Identity mismatch, ethical misuse	One
DAVS [101] (2019)	GAN	LRW	Enhances lip realism and intelligibility	Speed and variability limitations	One
LMGG [102] (2018)	GAN	GRID [57], LDC, LRW	Cross-modal speech-lip fusion	Realism + scale lacking, costly	Multi
VisemeNet [103] (2018)	LSTM	GRID [57], SAVEE, BIWI 3D	LSTM viseme curves improve sync robustness	emotion diversity is limited	Multi
FEP [104] (2017)	HMM	Custom Dataset	HMM + DNN with facial embeddings for pixel sync	Needs large training; overfitting risk	One
Facial Reenact. [10] (1989)	GAN	CelebA	cGAN + RNN for audio-face sync	Speaker variability, weak realism	Multi

aware geometric encoders, enabling cross-lingual synthesis and temporally stable lip-viseme alignment—forming the foundation for multilingual-ready architectures.

2.3 Video-based THG

Video-driven talking head generation (THG) focuses on synthesizing realistic videos by transferring motion from a driving sequence while preserving the target identity and ensuring temporal coherence. Figure 6 illustrates a generalized video-based THG framework combining audio, visual

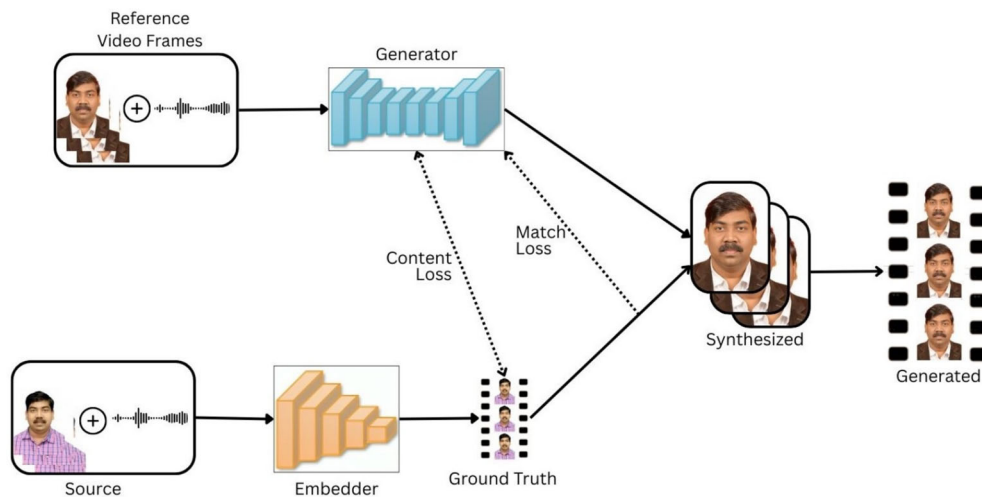


Fig. 6 Generalized approach for video-based

identity, and reference posture for synchronized head and facial motion generation.

As depicted in Fig. 6, the framework employs an Embedder to extract phonetic cues and a Generator to produce video frames, synchronized with reference postures and audio inputs. A Discriminator ensures realism through adversarial learning, minimizing content loss for facial motion accuracy and match loss to align synthesized videos with ground truth data. Table 3 compares key video-based THG approaches, detailing architectures, datasets, highlights, and limitations.

As shown in Table 3, HiDe-NeRF [48] employs Neural Radiance Fields (NeRFs) integrated with 3D Morphable Models (3DMMs) to preserve identity under significant deformations, though occlusion handling remains challenging. DFA-NeRF [49] achieves high-fidelity synthesis but suffers from lengthy rendering times. Neural Talking-Head [119] uses adaptive instance normalization with temporal regularization for efficient bandwidth usage in video conferencing but introduces artifacts under extreme pose variations. DISCOHEAD [113] leverages a ResNet-18 backbone with a dense motion estimator to enhance expressive mouth region synthesis, improving lip-sync accuracy. MMVID [116] fuses multimodal inputs for coherent, diverse video synthesis, although it struggles with motion consistency. Face-Dubbing++ [30] facilitates multilingual dubbing with synchronized lip accuracy but faces prosody mismatches impacting realism. FT [120] introduces few-shot meta-learning for rapid identity adaptation but trades off between realism and generalization. Similarly, vid2vid [121] adapts video-to-video synthesis for novel identities but underperforms on unseen styles. Speech2Vid [122] synthesizes real-time lip-synced outputs from speech and image inputs but is sensitive to accents and wide emotional variations. AdaIN-based frameworks [123] improve style transfer in dubbing scenarios but have limited expression precision and

raise privacy concerns. Overall, video-based THG techniques demonstrate rapid progress toward photorealistic, temporally consistent synthesis. However, challenges persist in real-time processing, robust occlusion handling, and balancing high-resolution quality with computational efficiency. Integrating diffusion-based refinement and temporal transformers offers promising directions for enhancing video realism and motion coherence. Further, Fig. 3 highlights the transition from direct video-based reenactment with GAN-driven motion fields to volumetric 3D feature representations for view consistency. Subsequently, prosody-driven multilingual phoneme modeling and video dubbing strategies influenced unified frameworks combining 3D motion modeling with cross-modal lip synchronization.

2.4 Text-based THG

Text-driven talking head generation (THG) converts textual prompts into synchronized speech-driven facial animations by combining emotion modeling, phoneme prediction, and visual synthesis. Figure 7 shows a generalized framework where a pre-trained model integrates textual inputs, visual identity references, and speech motion cues to generate high-fidelity videos.

As illustrated in Fig. 7, the framework uses an Embedder to extract phonetic and prosodic features from text-driven audio, while a Generator synthesizes temporally consistent video frames aligned with speech. The Discriminator ensures photorealism, guided by content loss for motion and lip-sync precision and match loss for realism. Beyond text-driven synthesis, cross-modal drivers are emerging as a transformative direction for next-generation THG systems. Integration of haptic feedback allows models to encode tactile emotional cues into facial expression synthesis which modulates micro-expressions based on real-time touch-derived affective states.

Table 3 Comparison of video-based approaches

Method	Arch.	Dataset	Highlights	Limitations	N-shot
DISCOHEAD [113] (2025)	ResNet-18	Obama, GRID [57], KoEBA	Dense motion estimator with encoder for expressive mouth region synthesis	Needs high-quality input; struggles with complex backgrounds	Multi
X-NeMo [114] (2025)	U-Net	HDTF [24], VFHQ [87]	Reenactment of expressive neural motion using disentangled latent attention	Face difficulties using exaggerated expressions that aren't included in the training data	Multi
Follow-Your-Emoji [115] (2024)	GAN	Custom Dataset	Expressive and Finely Controllable Freestyle Portrait Animation	Struggles with maintaining identity consistency and handling extreme facial expressions during freestyle portrait animation	Multi
HiDe-NeRF [48] (2023)	3DMM	VoxCeleb1 [55], TH-1KH [19]	Preserves identity under deformation for 3D face synthesis	Occlusion challenges; DeepFake misuse risk	Multi
MMVID [116] (2023)	GAN	MUG, iPER, VoxCeleb [17]	Fuses visual modalities for diverse generation	Motion consistency still limited	Multi
DFA-NeRF [49] (2022)	NeRF	LRS2 [117], HDTF [24]	Long rendering time; diarization dependency	Multi	
Face-Dubbing++ [30] (2022)	GAN	LRS2 [117]	Multilingual dubbing with sync accuracy	Prosody and ASR mismatch hurt realism	Multi
ROME [118] (2022)	FLAME	VoxCeleb2 [18], H3DS	One-shot head reenactment with realism	Lacks fine detail; smoothed geometry	One
Neural TH [119] (2021)	GAN	VoxCeleb2 [18], TH-1KH [19]	Bandwidth-efficient synthesis with unsupervised 3D keypoints	Prone to artifacts under pose variation	Multi
RHM [27] (2020)	GAN	CREMA-D [20], LRS3-TED [29]	3D-aware hybrid embedding for photorealism	Complex design, scalability issues	Multi
FT [120] (2019)	VGG19	VoxCeleb1 [55], VoxCeleb2 [18]	Few-shot generation with meta-learning init	Trade-off in realism and generalization	Multi
vid2vid [121] (2019)	GAN	YouTube, Street-scene	Few-shot with adaptive weights	Fails with unseen CG styles	Multi
Speech2Vid [122] (2017)	CNN	VoxCeleb [17], LRW	Real-time video generation from speech + image	Weakness with accents and wide expressions	Multi
AdaIN [123] (2017)	STN	RAVDESS [81]	Temporal regularized style transfer for dubbing	Low expression precision, privacy risks	Multi

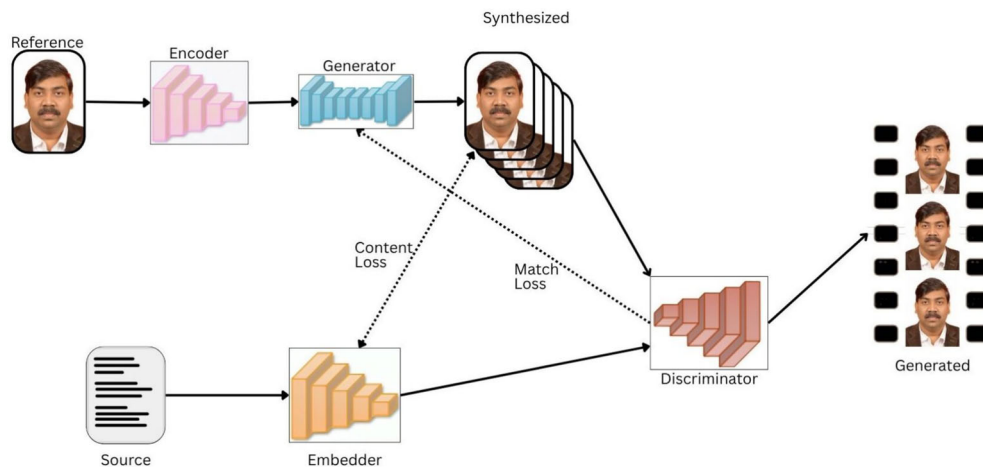


Fig. 7 Generalized approach for text-based

Similarly, brain–computer interface (BCI) pipelines leverage non-invasive EEG decoding to directly map neural intent into viseme sequences, enabling speech-free avatar control and assistive communication for patients with motor impairments. Future multimodal fusion frameworks are expected to unify text, audio, haptics, and BCI signals within a shared latent space, leveraging cross-modal embeddings to capture semantic intent, prosody dynamics, and affective depth. Such hybrid pipelines have vast applications, including immersive telepresence, cross-lingual virtual tutoring, empathy-driven digital assistants, and neuroadaptive avatar generation. These developments mark a paradigm shift from purely text-driven synthesis toward biologically grounded, multimodal THG systems, redefining naturalistic interaction fidelity. Table 4 summarizes state-of-the-art text-based THG techniques, including architectures, datasets, highlights, and limitations.

As shown in Table 4, InstructAvatar [124] produces interactive avatars from text, though emotional granularity remains limited. TalkCLIP [125] uses CLIP-based modulation to link semantic text and video synthesis but faces challenges with abstract expressions. FT2TF [126] integrates text and vision fusion for state-of-the-art talking head generation but suffers from high training costs and limited expression control. DiffTalk [12] introduces diffusion-based modeling for coherent motion synthesis but relies heavily on reference images. Face-Dubbing++ [30] preserves voice characteristics during multilingual dubbing but struggles with prosody mismatches. GC-AVT [98] offers fine-grained control over pose, lip movements, and facial expressions but generates inconsistent backgrounds. Text2Video [128] maps phoneme-to-pose alignment directly from text but lacks realism in dynamic scenarios. Write-a-Speaker [129] generates rhythmic animations from text, incorporating emotion and timing control, but requires motion capture data. Audio2Head [100] predicts dynamic head poses from text-

driven audio but suffers from identity mismatches in complex scenarios. LMGG [102] integrates cross-modal inputs for lip-sync consistency but lacks photorealism and processing efficiency. Recent advancements in text-conditioned THG include GenCA [136], which enables fine-grained semantic control of facial dynamics through textual prompts, and T3M [137], which extends generation into 3D space to synthesize identity-consistent avatars from natural language descriptions. STAR [138] pushes further into 4D synthesis by leveraging skeleton-aware text-driven motion for temporally coherent dynamic avatars. Text-to-Video [139] introduces a two-stage zero-shot framework mapping text directly to video sequences, enabling generalized talking head generation without speaker-specific training. Collectively, text-based THG has evolved into an expressive, controllable paradigm, enabling high-fidelity avatar synthesis from semantic inputs. However, challenges remain in emotional expressiveness, dynamic realism, and computational efficiency, suggesting future integration of diffusion-based refinement and multimodal transformers for improved synthesis quality. As summarized in Fig. 3, text-driven THG started with models establishing semantic text alignment but lacked expressive controllability. Recent diffusion-driven frameworks integrate phoneme-conditioned transformers to achieve synchronized expression, pose, and lip motion from textual prompts, later extended for multilingual, instruction-guided avatar generation.

2.5 2D-based THG

2D-based talking head generation (THG) employs landmark-driven warping and attention mechanisms to generate facial animations from limited visual input, prioritizing computational efficiency. These techniques typically map facial landmarks and expressions onto static images or video frames to produce motion. Table 5 compares state-of-the-art 2D

Table 4 Comparison of text-based approaches

Method	Arch.	Dataset	Highlights	Limitations	N-shot
InstructAvatar [124] (2024)	VAE	MEAD [22]	Text-guided expressive 2D avatars with improved interactivity	emotion granularity and text precision need work	One
EMO [85] (2024)	VAE	CREMA	Realism via dynamic audio-facial alignment	Data quality, expression complexity	One
TalkCLIP [125] (2024)	T2SS	MEAD [22], HDTF [24], VoxCeleb2 [18]	Text-driven synthesis via CLIP modulation	Abstract expressions, mouth artifacts	One
FT2TF [126] (2024)	GAN	LRS2 [117], LRS3	Text + vision fusion yields SOTA talking faces	High training cost, expressiveness gaps	Multi
DiffTalk [12] (2023)	GAN	HDTF [24]	Denoising diffusion for coherent motion	Relies on reference image, heavy compute	Multi
Wav2Lip [96] (2023)	LSTM	LRS2	Maintain accurate lip synchronization	Computational cost for outputs may be higher	Multi
StyleTalk [127] (2023)	3DMM	HDTF [24]	Controllable Speaking Styles Driven by Images	Difficulties in preserving identity and achieving accurate style control in diverse scenarios	One
Face-Dubbing++ [30] (2022)	LSTM	LRS2 [117], TED, etc	Voice-preserving dubbing with sync accuracy	ASR + prosody mismatches, fragility	Multi
GC-AVT [98] (2022)	GAN	VoxCeleb2 [18], MEAD [22]	Fine-grained pose + lip + expression control	Limited realism from background masking	Multi
Text2Video [128] (2022)	GAN	VidTIMIT	Phoneme-pose mapping for text-driven synthesis	Lacks realism, generalization, speaker control	Multi

Table 4 continued

Method	Arch.	Dataset	Highlights	Limitations	N-shot
Write-a-Speaker [129] (2021)	3DMM	Mocap	Text to head animation with rhythm and emotion control	Motion capture needed; performance cost	Multi
Flow Guided [24] (2021)	3DMM	HDTF	Flow-guided video from audio-text inputs	Poor coherence, style inconsistency	One
PC-AVS [130] (2021)	GAN	VoxCeleb2 [18], LRW	Pose control + lip-sync from raw face image input	Training bias; not fully real-time	One
Audio2Head [100] (2021)	RNN	LRW, GRID [57], VoxCeleb [17]	Audio-driven with pose + motion field prediction	Misuse potential, identity mismatch	One
MakeItTalk [131] (2020)	AUTOVC	VoxCeleb2 [18]	Expression-aware synthesis from one image + audio	Sparse landmarks may distort; lacks 3D cues	One
TFGAN [132] (2020)	GAN	Tacotron2 [133]	Domain based on time and frequency for high-fidelity speech synthesis	High computational complexity and challenges in maintaining naturalness	Multi
LMGG [102] (2018)	CNN	GRID [57], LDC, LRW	Cross-modal synced lip movement generation	Weak realism + efficiency	Multi
Facial Reenactment [134] (2018)	GAN	CelebA	RNN + cGAN for expressive face sync	Still lacks realism, consistency	Multi
ObamaNet [135] (2017)	LSTM	Char2Wav, Pix2Pix	Full pipeline: text to lip-sync video	Language support + ethics safeguards needed	Multi
Speech2Vid [122] (2017)	CNN	VoxCeleb [17], LRW	Real-time generation from audio + image	Accent handling, realism issues	One

Table 5 Comparison of 2D-based approaches

Method	Arch.	Dataset	Highlights	Limitations	N-shot
Style Transfer [140] (2024)	ResNet50	VoxCeleb2 [18], RAVDESS [81]	2D animation using audio and style reference from one image	Needs style frames; heavy computation	Multi
Diffused Heads [141] (2023)	GAN	LRW, CREMA	Autoregressive diffusion for smooth expressive motion	Too slow for real-time	Multi
MetaPortrait [35] (2023)	3DMM	VoxCeleb2 [18], HDTF [24]	Identity preservation using dense landmarks and generative priors	Blurring under occlusion; alpha-blend needed	Multi
LSP [95] (2021)	LSTM	Custom Dataset	Real-time facial animation from live speech	Affected by lighting, emotion, tracking	Multi
PC-AVS [130] (2021)	GAN	VoxCeleb2 [18], LRW	Pose control + lip-sync from raw face image input	Training bias; not fully real-time	One
MakeItTalk [131] (2020)	AUTOVC	VoxCeleb2 [18]	Expression-aware synthesis from one image + audio	Sparse landmarks may distort; lacks 3D cues	One
Video Rewrite [142] (1997)	Beier-Neely	HMM, TIMIT	Classic method using phoneme-aligned mouth patches	Weak for expressive motion; limited matching	One

approaches in terms of architecture, datasets, highlights, and limitations.

As shown in Table 5, Style Transfer [140] generates 2D animations by applying visual style features extracted via ResNet50 from a single image and audio input but requires extensive style reference data and high computational resources. Diffused Heads [141] integrates autoregressive diffusion into GAN architectures to produce smooth, expressive talking heads but lacks real-time performance. MetaPortrait [35] introduces a one-shot framework combining dense landmarks, 3D convolution-based temporal modeling, and generative priors to achieve high-fidelity identity preservation, evaluated on VoxCeleb2 [18] and HDTF [24] datasets. LSP [95] supports live speech-driven portrait animations using deep learning for dynamic head motion control but struggles under fast speech, expressive tones, and inconsistent lighting. PC-AVS [130] employs GANs with implicit pose codes to control head orientation and lip synchronization from raw face images, showing strong performance on VoxCeleb2 and LRW datasets but suffers from training bias and limited real-time adaptability. MakeItTalk [131] uses AUTOVC for one-shot expressive lip-sync generation but is sensitive to sparse landmarks and lacks explicit 3D awareness, reducing output quality in complex scenarios. Video Rewrite [142], an early phoneme-aligned technique, uses Hidden Markov Models and Beier-Neely warping to match speech-driven lip shapes to source videos. While effective for basic dubbing, it struggles with emotion-rich scenarios and complex expressions. Collectively, 2D-based THG demonstrates efficient and identity-preserving synthesis but faces challenges in real-time rendering, landmark robustness, and handling large pose variations. Future work may integrate

diffusion-based refinement and hybrid 2D-3D modeling for improved quality and temporal coherence. Along with that, Fig. 3 illustrates the methodological leap from 2D geometric priors to explicit 3D head pose reconstruction. This transition resolved view-inconsistency challenges and informed the design of unified frameworks combining multi-view geometric consistency with generative diffusion-based temporal refinement.

2.6 3D-based THG

3D-based talking head generation (THG) techniques leverage morphable models and neural rendering to achieve realistic synthesis by disentangling expression, posture, and geometry. Figure 8 illustrates a generalized architecture where a Generator synthesizes identity-preserving 3D head models from source images and landmark sequences, while a Discriminator refines realism via adversarial learning.

Table 6 summarizes notable 3D-based approaches, highlighting architectures, datasets, contributions, and limitations.

As shown in Table 6, PACT [143] employs 3D U-Nets to correct skull distortion in reconstruction but struggles with anatomical shape variance. AniArtAvatar [144] generates artistic avatars using 3D Morphable Models (3DMMs), SDF-based neural surface representation, and diffusion-based textures, enabling body and face motion control but suffers from style inconsistencies. JambaTalk [145] integrates VOCA [146], FaceFormer, and FaceDiffuser [153] to model speech and laughter via a two-stage pipeline but lacks comprehensive gesture modeling. ADL [147] supports multilingual lip-sync using Wav2Vec2.0 with transformer-based

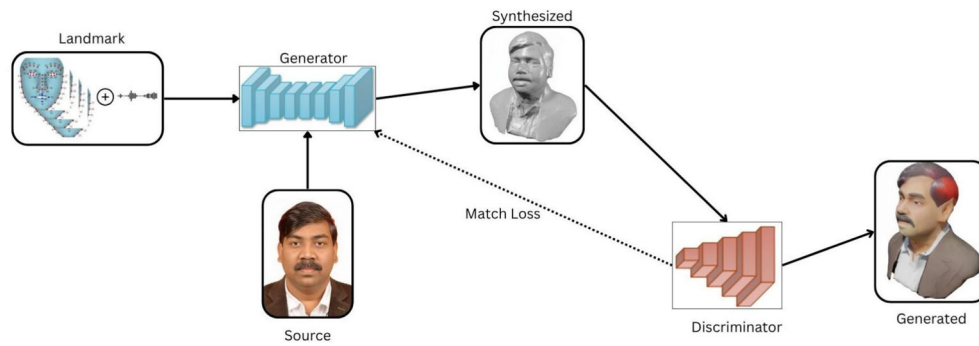


Fig. 8 Generalized approach for 3D-based

Table 6 Comparison of 3D-based approaches

Method	Arch.	Dataset	Highlights	Limitations	N-shot
PACT [143] (2025)	3D U-Net	Custom Dataset	Overcomes skull distortion via deep reconstruction	Sensitive to skull shape variance	One
AniArtAvatar [144] (2025)	3DMM	Wonder3D, NeuS	Artistic avatar generation with body + face motion	Style deformation and inconsistency	One
JambaTalk [145] (2024)	Mamba	VOCA [146], FaceFormer	Two-stage modeling for speech + laughter	emotion/gesture modeling missing	Multi
ADL [147] (2024)	OM-Net	VOCA [146]SET-lip, ADLSET	Language-agnostic lip-sync with Wav2Vec2.0	Ignores upper-face emotion dynamics	Multi
FHP [148] (2024)	FFNN	StateFarm, Pose	Posture detection in real-time use cases	Poor cross-domain generalization	One
TTSF [149] (2024)	3DMM	LRS3, VoxCeleb2 [18]	TTS fused with facial generation for avatars	Voice consistency and realism gaps	One
LaughTalk [150] (2023)	3DMM	VoxCeleb2 [18], CelebV-HQ [25]	Expressive generation for laughter + speech	Weakness under emotional extremes	Multi
AE-NeRF [151] (2023)	3DMM	NeRF, HDTF [24]	Audio-enhanced NeRF for sync + fidelity	Needs better speed benchmarking	One
PV3D [47] (2023)	GAN	VoxCeleb [17], CelebV-HQ [25]	3D-consistent motion with editable views	Needs more diverse training data	Multi
DVP [152] (2018)	cGAN	Leader Dataset	Transfers expression, pose, blink for reenactment	Breaks under extreme motion	Multi

audio embeddings, achieving accurate visual alignment but ignoring upper-face emotion dynamics. FHP [148] focuses on 3D posture detection for real-time applications but struggles with cross-domain generalization. TTSF [149] combines text-to-speech and facial synthesis for synchronized avatars but faces challenges in maintaining natural voice consistency. LaughTalk [150] incorporates FLAME [154] parameters to jointly model speech and laughter but performs poorly under high-intensity emotions. AE-NeRF [151] combines Neural Radiance Fields (NeRFs) with 3DMMs for audio-driven synchronization and view-consistent outputs, though speed optimization remains limited. PV3D [47] leverages GANs with motion dynamics for 3D-consistent portrait videos but requires more diverse training data. Deep Video Portraits (DVP) [152] introduced conditional GAN-based pose and expression transfer, enabling realistic eye-blink and head motion synthesis but struggles under extreme motion and

occlusions. Additionally, part-based fusion strategies such as PGN-PFN [155] enhance robustness to occlusions while preserving identity attributes through localized 3D feature blending. Overall, 3D-based THG provides controllable, high-fidelity modeling by explicitly capturing geometry, emotion, and pose dynamics. However, challenges remain in handling emotional diversity, occlusions, computational efficiency, and cross-domain adaptability. Future research may integrate NeRF-based free-view synthesis, multilingual audio-visual modeling, and hybrid diffusion frameworks for scalable and photorealistic 3D avatar generation. As depicted in Fig. 3, 3D Morphable Models underpinned early parametric THG pipelines but were limited by rigid mesh templates. Advancements in mesh deformation enabled flexible vertex-level motion control, inspiring volumetric pipelines and culminating in unified frameworks integrating 3D geometry, radiance fields, and diffusion-based refinement.

2.7 Neural Radiance Fields (NeRF)-based THG

Neural Radiance Fields (NeRFs) have revolutionized talking head generation (THG) by enabling photorealistic novel-view synthesis and dynamic facial deformation using volumetric scene representations. Table 7 compares major NeRF-based THG approaches, including their architectures, datasets, highlights, and limitations.

As shown in Table 7, CVTHead [156] introduces a one-shot controllable neural avatar framework leveraging 3D Morphable Models (3DMMs) for fine-grained pose and expression control but is limited by DECA-based mesh accuracy. Head3D [157] employs ConvLSTMs for interpretable motion transfer across driving videos, producing temporally coherent canonical frames but faces parsing errors and difficulties with side-view consistency. SSP-NeRF [158] integrates semantic priors into NeRF for audio-driven dynamic sampling, improving lip-sync fidelity but suffers from slow rendering and weak generalization. IMavatar [159] learns implicit head avatars from monocular videos, modeling geometry and expressions robustly, though high computational costs and limited high-frequency details remain challenges. ROME [118] performs one-shot mesh-based head reenactment by regressing FLAME parameters from a single image, enabling fast geometry recovery but smoothing fine facial details. FNeVR [160] improves motion deformation handling through neural volume rendering but faces performance bottlenecks due to high computation demands and restricted pose editability. DFA-NeRF [49] combines NeRF with 3DMM-based audio conditioning to achieve high-fidelity, personalized talking heads but lacks multi-speaker support and suffers from slow rendering times. AD-NeRF [88] advances view- and audio-adaptive synthesis but encounters realism degradation in cross-lingual scenarios. NerFACE [161] employs dynamic 4D NeRF modeling for telepresence applications but omits torso and eye-motion controls. HeadNeRF [67] combines NeRF with 3DMMs to model fine-grained head geometry, but its generalization fails under complex headgear or extreme motion. Emerging methods like NeRFInverter [162] bridge NeRF and GANs for high-resolution volumetric inversion, enabling controllable fine-tuning. Similarly, DFRF [163] captures dynamic appearance and expression changes, enhancing temporal coherence. Additionally, frameworks like 3DFaceShop [164] enable sketch-based editing of 3D-aware talking heads, while Next3D [165] employs generative neural textures to synthesize high-quality, animatable facial avatars with photorealistic fidelity. Overall, NeRF-based THG advances high-fidelity, controllable, and identity-preserving synthesis. However, real-time rendering, multi-view consistency, computational efficiency, and large-scale data requirements remain significant challenges. Future directions include integrating hash-encoded NeRF optimizations, hybrid GAN-

NeRF pipelines, and diffusion-based refinement to improve temporal stability and scalability. As shown in Fig. 3, NeRF-based models introduced volumetric radiance field representations for free-view synthesis and improved identity consistency under extreme pose changes. Their evolution into Gaussian splatting-enhanced pipelines and integration with diffusion priors informed unified frameworks enabling multi-view realism, pose robustness, and low-latency rendering.

2.8 Diffusion-based THG

Diffusion models have recently advanced talking head generation (THG) by achieving highly realistic temporal coherence, smooth facial expression synthesis, and controllable emotional dynamics. Table 8 summarizes major diffusion-based THG approaches, their architectures, datasets, highlights, and limitations.

As seen in Table 8, MoDiTalker [166] introduces a motion-disentangled diffusion model that separately predicts lip and non-lip facial dynamics, improving synchronization and naturalness but requiring significant compute and long sampling times. DreamTalk [167] employs a style-aware diffusion framework combining a style predictor, style-driven lip expert, and denoiser to synthesize emotion-controllable talking heads, though subtle emotion generalization remains limited. EMOTALKER [168] incorporates CLIP-based embeddings to provide fine-grained control over emotion intensity in speech-driven avatars but faces trade-offs between realism and expressiveness. DiffTalk [12] leverages RNN-guided denoising diffusion for temporally consistent audio-driven talking heads but depends on reference images and struggles with slow inference. Diffused Heads [141] integrates autoregressive diffusion into GANs to generate smooth expressive motion, achieving high quality on CREMA and LRW datasets but unsuitable for real-time synthesis. Diffusion Video Editing [169] combines U-Net architectures with denoising diffusion to edit multi-speaker videos using speech input, but extended training times and lip-sync inconsistencies remain challenges. DAE-Talker [170] applies a denoising autoencoder within latent diffusion to improve lip-sync precision, pose control, and expressiveness, yet it suffers from high resource demands and limited generalization across unseen identities. DiT-Head [171] leverages diffusion transformers for high-resolution synthesis with strong temporal fidelity but faces heavy computational costs and English-only dataset limitations. FaceDiffuser [153] learns 3D speech-driven facial motions via stochastic diffusion on BIWI, Multiface, and BEAT datasets, producing natural deformations but limited by slow inference and dataset diversity. PAVDP [172] incorporates probabilistic priors into one-shot diffusion-based synthesis, enhancing holistic expression generation but introducing

Table 7 Comparison of NeRF-based approaches

Method	Arch.	Dataset	Highlights	Limitations	N-shot
CVTHead [156] (2023)	3DMM	VoxCeleb [17]	One-shot neural avatar w/ expression + pose control	Limited by DECA mesh reconstruction	One
Head3D [157] (2023)	ConvLSTM	VoxCeleb [17], FaceForensics	Canonical frames w/ interpretable 3D motion transfer	Parsing errors, side-view issues	Multi
SSP-NeRF [158] (2022)	3DMM	NVP, Obama Synth	Semantic NeRF w/ dynamic sampling	Rendering slow; poor language generalization	Multi
IMavatar [159] (2022)	3DMM	VOCA [146], COMA	Monocular avatars w/ expression modeling	Heavy compute, coarse high-frequency detail	Multi
ROME [118] (2022)	FLAME	VoxCeleb2 [18], H3DS	One-shot head reenactment with realism	Lacks fine detail; smoothed geometry	One
FNeVR [160] (2022)	CNN	VoxCeleb1 [55]	Volume rendering to fix motion deformation	Compute-heavy; poor pose editability	One
DFA-NeRF [49] (2022)	3DMM	LRS2 [117], HDTF [24]	Audio-driven NeRF w/ lip-sync alignment	Single speaker support; slow rendering	Multi
AD-NeRF [88] (2021)	GAN	Wild videos	View- and audio-adaptive NeRF synthesis	Language mismatch degrades realism	Multi
NerFACE [161] (2020)	SRNs	Wild videos	4D NeRF reenactment for telepresence	Lacks eye/torso motion control	Multi
HeadNeRF [67] (2019)	3DMM	CelebAMask-HQ, FFHQ [23]	Fine-grain NeRF face model	Fails on headgear, wild generalization	Multi

visual artifacts in high-frequency details. The Adaptive Diffusion Landmark Dynamic Rendering framework (DLDR) [173] further refines synchronization accuracy by integrating transformer-based audio-semantic mapping and landmark alignment, enhancing perceptual quality and realism. Overall, diffusion-based THG has significantly advanced lip-sync accuracy, emotional expressiveness, and temporal coherence. However, long sampling times, VRAM requirements, computational costs, and potential ethical misuse remain challenges. Future directions may integrate hybrid diffusion-GAN pipelines, real-time landmark conditioning, and multimodal reinforcement learning to improve scalability and control. Additionally, Fig. 3 highlights the shift from GAN-only pipelines to diffusion-driven temporal stabilization. Recent techniques addressed temporal flickering, cross-identity adaptation, and multi-expression alignment, with their integration into volumetric priors directly influencing

hybrid designs combining diffusion-driven synchronization with 3D-aware photorealistic rendering.

2.9 3D animation-based THG

3D animation-based talking head generation (THG) integrates motion capture, physics-driven simulation, and rigged 3D models to achieve expressive and controllable avatars. These techniques produce high-fidelity visualizations for VFX, gaming, and virtual telepresence. Table 9 compares major 3D animation-based THG methods, highlighting architectures, datasets, contributions, and limitations.

As shown in Table 9, MultiTalk [174] uses a VQ-VAE architecture with multilingual embeddings to generate expressive avatars but struggles with non-standard pronunciations. CSTalk [175] integrates temporal convolutional networks (TCNs) to synchronize lips and emotions but han-

Table 8 Comparison of diffusion-based approaches

Method	Arch.	Dataset	Highlights	Limitations	N-shot
MoDiTalker [166] (2024)	GAN	LRS3-TED [29], HDTF [24]	Motion-disentangled diffusion for subtle lip movements	High sampling time, stochasticity, ethics concerns	Multi
DreamTalk [167] (2024)	3DMM	MEAD [22], HDTF [24]	Style-aware emotional diffusion model	Lacks realism for subtle emotional cues	Multi
EMOTALKER [168] (2024)	CLIP	MEAD [22], CREMA-D [20]	Emotion intensity control in speech-driven avatars	Generalization issues, realism-emotion trade-offs	One
DiffTalk [12] (2023)	RNN	HDTF [24]	Audio-driven diffusion with temporal coherence	Requires face reference image, slow, artifacts	Multi
Diffused Heads [141] (2023)	GAN	CREMA, LRW	Autoregressive diffusion for smooth expressions	Slow generation; not real-time suitable	One
Diffusion Video [169] (2023)	U-Net	GRID [57], CREMA-D [20]	Denosing diffusion for multi-speaker video edits	Long training, lip-sync issues	One
DAE-Talker [170] (2023)	DAE	LibriTTS	Latent DAE boosts lip-sync and pose accuracy	Poor pose modeling; resource intensive	One
DiT-Head [171] (2023)	DiT	HDTF [24]	High-fidelity transformer-based diffusion	English-only; deepfake + compute risks	Multi
FaceDiffuser [153] (2023)	HuBERT	BIWI, Multiface, BEAT	3D animation via nondeterministic diffusion	Dataset limits, slow inference, subjective evals	Multi
PAVDP [172] (2022)	3DMM	VoxCeleb1 [55]	Probabilistic priors for facial motion	Stripe artifacts; weak high-freq detail	One

Table 9 Comparison of 3D animation-based approaches

Method	Arch.	Dataset	Highlights	Limitations	N-shot
MultiTalk [174] (2024)	VQ-VAE	Custom Dataset	Multilingual avatars with style embeddings for lip-sync	Language limits and non-standard speech issues	Multi
CSTalk [175] (2024)	TCN	Live Link Face	3D lip + Emotion sync aligned to speech	Narrow emotion range; complex cues poorly handled	Multi
Learn2Talk [176] (2024)	TCN	BIWI, VOCASET [146]	2D generation fused with 3D for clarity	Weak realism in blinking/emotion; multitasking issues	Multi
DiffSpeaker [177] (2024)	GRU	BIWI, VOCASET [146]	Fast transformer-based 3D facial motion	Struggles with realism + 4D audio mapping	Multi
Media2Face [178] (2024)	VAE	BIWI, VOCASET [146]	Diffusion + parametric asset + dataset fusion	Compute-heavy; dataset-dependent	Multi
PMMTalk [179] (2023)	CNN	3D-CAVFA, VOCASET [146]	Pseudo-multimodal lip-blendshape sync	No head/exp control; inference slow	Multi
3DiFACE [180] (2023)	Transformer	VOCASET [146]	Personalized diffusion-based 3D animation	Small dataset, editing limits	Multi
Speech4Mesh [181] (2023)	FLAME	MEAD [22], VoxCeleb2 [18]	Audio2Mesh system handles 4D mesh scarcity	Prone to overfitting, poor generalization	One
PV3D [47] (2023)	GAN	VoxCeleb [17], CelebV-HQ [25], TH-1KH [19]	GAN + motion modeling for 3D-aware output	Quality limited by 2D data	One
3D-TalkEmo [182] (2021)	StarGAN	Custom Dataset	Emotion-adaptive 3D talking head generation	Low scalability; data/tuning bottlenecks	Multi

dles a limited range of affective cues. Learn2Talk [176] fuses 2D and 3D synthesis for improved lip-sync and vertex accuracy but underperforms in secondary expression realism. DiffSpeaker [177] employs GRU-driven parallel generation for faster speech-to-3D animation but struggles with accurate 4D emotion mapping. Media2Face [178] integrates diffusion modeling with parametric face assets to achieve expressive animations, although computational cost and dataset dependence limit scalability. PMMTalk [179] adopts CNN-based pseudo-multimodal mappings for blendshape synchronization but lacks controllability over head pose and fine-grained expressions. 3DiFACE [180] introduces a diffusion-based personalized framework using transformer-driven audio conditioning, enabling stochastic 3D animation editing but constrained by small dataset sizes. Speech4Mesh [181] leverages FLAME [154] models to predict mesh-based speech deformations, offering robust geometry modeling yet prone to overfitting and poor cross-domain generalization. PV3D [47] combines GAN-based generation with explicit motion dynamics for 3D-aware portrait videos but is limited by insufficient 2D data quality. 3D-TalkEmo [182] employs a StarGAN architecture for multilingual emotional animation, improving lip alignment and emotional expression but facing scalability and tuning challenges. Overall, as summarized in Table 9, 3D animation-based THG provides high expressiveness, multilingual adaptability, and fine-grained control over mesh-based avatars. However, challenges remain in dataset

scarcity, realistic secondary expressions, and efficient multimodal integration. Future research may leverage hybrid diffusion transformer pipelines, differentiable physics for dynamic cloth and hair modeling, and cross-lingual embeddings to enhance realism and controllability. As summarized in Fig. 3, hybrid models pioneered multilingual dubbing and prosody-aware synchronization, unifying Gaussian-based shape modeling with volumetric rendering. These innovations culminate in integrated frameworks combining 3D motion dynamics, diffusion-stabilized multi-view consistency, and cross-lingual expressivity.

2.10 Real-time rendering (RTR)-based THG

Real-time rendering (RTR)-based talking head generation (THG) represents a significant advancement over conventional image-based and GAN-driven approaches by incorporating neural radiance fields (NeRFs), diffusion transformers, U-Nets, and 3D Gaussian splatting techniques. Unlike image-based frameworks that primarily rely on 2D motion transfer, RTR-based methods leverage implicit 3D scene representations to achieve photorealistic, temporally coherent, and multi-view-consistent facial animation. These approaches enable highly realistic talking head synthesis with minimal latency, supporting real-time interactive applications.

Recent developments in RTR-based THG focus on improving synthesis quality, maintaining identity consistency, and achieving real-time performance while generalizing across diverse subjects. State-of-the-art methods integrate advanced neural rendering backbones with audio- and text-conditioned motion control, enabling synchronized lip movements, natural facial expressions, and smooth pose transitions. Moreover, the adoption of 3D Gaussian splatting and transformer-driven diffusion models has significantly enhanced rendering speed and fidelity, paving the way for high-resolution, real-time portrait generation.

As referred in Table 10, **Ditto** [183] introduces a diffusion transformer (DiT) architecture for controllable real-time THG, achieving approximately **25 FPS**, though its frame rate remains lower than other recent methods. **OmniTalker** [184] extends DiT with joint modeling of speech and facial styles for synchronized text-driven generation but is limited to head-only rendering, potentially introducing boundary artifacts. **R2-Talker** [186] employs a NeRF-based progressive multilayer conditioning strategy to achieve highly realistic outputs at **32.44 FPS**, although it struggles to generalize effectively across diverse identities. SyncAnimation [188] adopts a U-Net-based framework optimized for precise pose control, achieving real-time avatar generation at **41 FPS**, but its performance degrades when capturing fine facial details. Similarly, ChatAnyone [189] uses a GAN-driven hierarchical motion model for stylized portraits at **41 FPS**, yet its adaptability to diverse visual styles remains limited. GaussianSpeech [190] and GaussianTalker [191] pioneer the use of 3D Gaussian splatting for photorealistic THG. While GaussianTalker achieves up to **120 FPS**, offering the highest rendering speed among surveyed models, it suffers from high preparation costs and reduced generalization to novel identities. Hallo2 [192] and EchoMimic [193] focus on high-resolution, long-duration portrait animation. Hallo2 [192] achieves seamless motion synthesis but may introduce artifacts due to its patch-drop data augmentation, whereas EchoMimic [193] relies on landmark-driven control but lacks a fully optimized video-specific architecture, limiting its efficiency. Furthermore, GaussianSplatting [194] generalizes real-time 3D scene synthesis at 1080p and 30 FPS, representing a significant step toward interactive rendering pipelines. Finally, Mip-NeRF 360 [195] extends NeRF-based representations to unbounded scene reconstruction, enabling realistic multi-view synthesis but encountering limitations with low-resolution outputs when training data are sparse. In summary, RTR-based THG approaches deliver substantial improvements in visual realism, frame rate, and interactivity compared to traditional GAN-driven frameworks. Gaussian splatting achieves state-of-the-art performance in speed and fidelity, while diffusion transformers and NeRF-based architectures excel in identity preservation and cross-modal synchronization. Future research is

expected to focus on integrating multimodal conditioning (speech, text, gaze, and gesture), enhancing generalization across unseen identities, and enabling photorealistic, real-time 3D avatars suitable for metaverse applications, AR/VR environments, and telepresence systems. Furthermore in Fig. 3, the overall innovation trajectory moves from local affine transformations to geometry-driven disentanglement, volumetric rendering, and diffusion-driven temporal stabilization, culminating in unified frameworks that deliver real-time, multi-view-consistent, multilingual-ready avatar synthesis.

3 Evaluation metrics taxonomy

Assessing talking head generation (THG) models requires a multifaceted approach that includes evaluating various dimensions of the output, including visual authenticity, harmony between audio and visual elements, and coherence of the conveyed message. Tools used in this field include dynamic image-based assessments, immersive audio-based criteria, insightful text-based analyses, and comprehensive video-based evaluations. Image-based metrics focus on evaluating the visual quality of generated frames, such as CSIM, SSIM, PSNR, PRMSE, L1 Loss, MKR, AUCON, AKD, SyncNet, FID, LMD, CPBD, AVD, and CLIPSIM. CSIM measures visual similarity by focusing on angular similarity between feature vectors of generated and real images, while SSIM focuses on structural information and perceptual quality. AUCON assesses the difference between processed and original pictures, while PRMSE indicates higher quality with lower error. AKD measures the alignment between predicted and actual facial key points, crucial for tasks like THG and facial expression generation. SyncNet evaluates the synchronization between audio and visual streams, focusing on lip synchronization and cross-modal temporal consistency. Video-based metrics are often used to evaluate generation models, such as FID (Fréchet Inception Distance), FVD (Fréchet Video Distance), E-FID (Efficient Fréchet Inception Distance), LMD (Learned Metric for Image Quality Assessment), CPBD (Cumulative Probability of Blur Detection), and AVD (Audio-Visual Discrepancy). Text-based metrics are relevant for evaluating semantic coherence when THG is driven by textual input. To evaluate THG progress objectively, researchers use a combination of metrics to understand model strengths and weaknesses, including visual quality, temporal coherence, audio-visual synchronization, and semantic accuracy. The development of more perceptually aligned evaluation metrics remains an important research area.

Table 10 Comparison of RTR-based approaches

Method	Arch.	Dataset	Highlights	Limitations
Ditto [183] (2025)	Diffusion Transformer (DiT)	HDTF [24], Talk9 [183]	Controllable real-time THG achieving around 25 FPS	FPS comparatively lower than similar models
OmniTalker [184] (2025)	DiT	SEED [185], VoxCeleb2 [18]	Real-time text-driven THG, jointly modeling speech and facial styles at 25 FPS	Focus on head-only generation leading to potential artifacts
R2-Talker [186] (2025)	NeRF	Geneface [187]	Realistic real-time THG progressive multilayer conditioning achieving 32.44 FPS	Limited generalization to diverse identities, affecting synthesis consistency
SyncAnimation [188] (2025)	U-Net	HDTF [24]	Enabling real-time avatar generation with precise pose achieving 41 FPS	Limitations in capturing fine facial details
ChatAnyone [189] (2025)	GAN	Custom Dataset	Stylized portrait using hierarchical motion model achieving 41 FPS	Limited adaptability to diverse styles and identities
GaussianSpeech [190] (2024)	FLAME [154]	Custom Dataset	Approach for high-fidelity photorealistic THG with facial details	Limiting to diverse accents, languages, and unseen facial variations
GaussianTalker [191] (2024)	3D Gaussian Splatting	HDTF [24]	Real-time high-fidelity synthesis with audio-driven achieving up to 120 FPS	Limited generalization to new identities, making preparation time-consuming
Hallo2 [192] (2024)	U-Net	HDTF [24], VoxCeleb [17]	Long-duration, high-resolution portrait animation for realistic and seamless facial motion	Introduce artifacts due to patch-drop data augmentation technique
EchoMimic [193] (2024)	U-Net	HDTF [24], CelebV-HQ [25]	Lifelike portrait animations enabled by editable landmark-based facial	Relies on adapted image techniques and lacks a true video processing architecture
GaussianSplatting [194] (2023)	ResNet	Mip-NeRF 360 [195]	Achieves 1080px synthesis with real-time rendering at 30 FPS	High computational demands and scalability challenges for complex scenes
Mip-NeRF 360 [195] (2021)	NeRF	Custom Dataset	Achieving realistic view synthesis, extends Mip-NeRF for unbounded scenes	Struggles with low-resolution renderings from limited images

3.1 Evaluation of image-based approaches

Based on the existing literature from sources [60] and [45], which were selected for comparison due to their relevant citations, we have produced a summary highlighting the comparison of various image-based approaches across different datasets and assessment metrics. This information can be found in Tables 11 and 12.

The evaluation of image-based THG models featured in the current literature is vividly encapsulated in Table. This comprehensive overview includes a diverse array of models-X2face [69], NeuralHead-FF [58], MarioNETte [67], FOMM [45], MeshG, OSFV [196], and DaGAN [60]-each subjected to rigorous scrutiny using standard metrics across two prominent datasets: VoxCeleb1 [55] and CelebV. The results definitively reveal that DaGAN [60] and OSFV [196] dominate the field, consistently delivering exceptional performance marked by impressive metrics in Content Similarity (CSIM), Peak Signal-to-Noise Ratio (PSNR), and Area Under the Curve of Object Consistency (AUCON). Their ability to produce strikingly realistic and coherent talking heads sets a benchmark for others. FOMM [45] and MeshG also shine, showcasing remarkable capabilities, especially in PSNR and CSIM.

In contrast, while X2face holds its own reasonably well, it pales compared to the prowess of its more recent counterparts. Meanwhile, NeuralHead-FF [58] struggles significantly, landing at the bottom of the performance spectrum among the models evaluated. When we turn our attention to the CelebV dataset, the formidable duo of OSFV [196] and DaGAN [60] continues to impress, with DaGAN [60] achieving the highest AUCON and PSNR, further complemented by the lowest Peak Root Mean Square Error (PRMSE). The tables also venture into motion-related metrics, evaluating similar THG models across various datasets, including the dynamic Tai-Chi-HD, the diverse VoxCeleb [17], the engaging EMO [85], and the intricate Bair. Here, FOMM [45] consistently excels, brilliantly capturing and replicating intricate motion dynamics, underlining its effectiveness and showcasing its superiority. Monkey-Net provides a commendable balance, standing strong in its performance, while X2face, regrettably, lags in motion metrics. These findings collectively highlight remarkable strides in the domain of THG. Models like DaGAN [60] and OSFV [196] achieve breathtaking visual quality and redefine the standards of realism. Meanwhile, Exceptional performance of FOMM [45] in motion accuracy draws attention to the ongoing evolution in this field. This review emphasizes these compelling results, delves into the intricate trade-offs between visual fidelity and motion realism, and proposes exciting avenues for future research and the development of new metrics.

3.2 Evaluation of audio-based approaches

A detailed comparison of diverse audio-based methodologies across various datasets and evaluation metrics is beautifully encapsulated in Tables 13 and 14. This insightful summary draws from the latest literature from [85] and [45], carefully chosen for their notable citations and significance within the scholarly landscape. Each methodology is meticulously evaluated, providing a comprehensive overview highlighting its strengths and weaknesses in tackling audio-related challenges.

The evaluation of various audio-driven THG models is thoroughly detailed in Table 2. Table 2 features a diverse array of models, including FOMM [45] (Few-Shot Video Generation), OSFV (One-Shot Face Video Generation) [196], DaGAN [60] (Dynamic Generative Adversarial Network), ROME [118] (Recurrent Oral Motion Encoder), FNeVR [160] (Fast Neural Video Representation), and HiDe-NeRF [48] (High-Definition Neural Radiance Field). These models' performance was assessed using three distinct datasets: VoxCeleb1 [55], VoxCeleb2 [18], and the more challenging TH-1KH [19].

The evaluation of video quality and synchronization involves metrics such as Content Similarity, AUCON, PRMSE, FID, and AVD. These measures assess the stability of facial features, pixel-level discrepancies, and alignment and synchronization of audio and visual components. They help determine the quality of generated content and the stability of facial features.

HiDe-NeRF [48] stands out as the leading model across all tested datasets. It achieves the highest scores for CSIM and AUCON, indicating top-tier content fidelity and consistency throughout the video. It minimizes PRMSE, FID, and AVD scores, underscoring its excellence in visual quality and audio-visual synchronization. ROME [118] also excels, particularly in the AVD metric, which suggests that it effectively captures the nuances of facial movements in concert with corresponding audio, enhancing the realism of the generated videos.

In terms of performance, OSFV [196] and DaGAN [60] exhibited competitive results. However, OSFV [196] generally surpassed DaGAN [60], indicating a more robust ability to generate convincing and coherent talking head videos. Notably, FOMM [45], while innovative, performed the most poorly among the newer models. FNeVR [160]'s outcomes place it in the middle range, showing potential yet lacking the statistical excellence of its counterparts.

The TH-1KH [19] dataset poses significant challenges to all models, as evidenced by the lower performance scores. This dataset likely contains complexities that test the limits of current generation techniques.

Overall, the findings presented in Table 2 illuminate the substantial progress made by recent models, especially HiDe-

Table 11 Evaluation of image-based (Part-1) [arrows indicate whether higher (↑) or lower (↓) values are better]

Dataset	Model	CSIM ↑	SSIM ↓	PSNR ↑	PRMSE ↓	AUCON ↑
VoxCeleb1	X2Face [69]	0.689	0.719	22.537	3.26	0.813
	NeuralHead-FF [58]	0.229	0.635	20.818	3.76	0.719
	MarioNETte [67]	0.755	0.744	23.244	3.13	0.825
	FOMM [45]	0.813	0.723	30.394	3.20	0.886
	MeshG [64]	0.822	0.739	30.394	3.20	0.887
	OSFV [196]	0.895	0.761	30.695	1.64	0.921
	DaGAN [60]	0.899	0.804	31.220	1.22	0.939
CelebV	X2Face [69]	0.450	–	3.620	–	0.679
	NeuralHead-FF [58]	0.108	–	3.300	–	0.722
	MarioNETte [67]	0.520	–	3.410	–	0.710
	FOMM [45]	0.462	–	3.900	–	0.667
	MeshG [64]	0.635	–	3.410	–	0.709
	OSFV [196]	0.791	–	3.150	–	0.805
	DaGAN [60]	0.723	–	2.330	–	0.873

Table 12 Evaluation of image-based (Part-2) [arrows indicate whether higher (↑) or lower (↓) values are better]

Dataset	Model	CSIM ↑	SSIM ↓	PSNR ↑	PRMSE ↓
Tai-Chi-HD	X2face [197]	0.08	17.654	0.109	0.272
	Monkey-Net [63]	0.077	10.798	0.059	0.228
	FOMM [45]	0.063	6.862	0.036	0.179
VoxCeleb	X2face [197]	0.078	7.687	–	0.405
	Monkey-Net [63]	0.049	1.878	–	0.199
	FOMM [45]	0.043	1.294	–	0.14
Nemo	X2face [197]	0.031	3.539	–	0.221
	Monkey-Net [63]	0.018	1.285	–	0.077
	FOMM [45]	0.016	1.119	–	0.048

Table 13 Evaluation of audio-based (Part-1) [arrows indicate whether higher (↑) or lower (↓) values are better]

Dataset	Model	FID ↓	SyncNet ↑	PRMSE ↓	F-SIM ↓	FVD ↓
HDTF	Wav2Lip [70]	9.38	5.79	80.34	407.93	0.693
	SadTalker [198]	10.31	4.82	84.56	214.98	0.503
	DreamTalk [167]	58.8	3.43	67.87	619.05	2.257
	MakeItTalk [131]	21.73	2.85	76.91	350.96	1.072
	EMO [85]	8.76	3.89	78.96	67.66	0.116

NeRF [48], in producing high-quality talking head videos with accurate synchronization between audio and visual elements. The analysis of the AVD metric reveals a notable performance gap among the various models, suggesting ample opportunity for further advancements in this field. The comprehensive review underscores the importance of meticulously evaluating visual quality and motion accuracy, highlights the challenges associated with different datasets, and emphasizes the continued significance of metrics such as AVD and AKD. Finally, it points to the necessity for consistent metric reporting to enhance comparability and advancement within THG.

3.3 Evaluation of video-based approaches

Based on the existing literature, which has been selected based on the citation in comparison with others, we have produced a summary of the comparison of different video-based approaches in different datasets across different assessment metrics in Tables 15 and 16.

The evaluation of video-driven THG models presented in Table includes FOMM [45], OSFV [196], DaGAN [60], ROME [118], FNeVR [160], and HiDe-NeRF [48], assessed across three datasets: VoxCeleb1 [55], VoxCeleb2 [18], and TH-1KH [19]. Key metrics used in this evaluation are CSIM (Content Similarity), AUCON (Area Under the Curve of

Table 14 Evaluation of audio-based (Part-2) [arrows indicate whether higher (↑) or lower (↓) values are better]

Dataset	Model	LMD↓	SSIM↑	PSNR↑	CPBD↑
GRID	Vondrick [199]	2.38	0.6	28.45	0.129
	Chung [122]	1.35	0.74	29.36	0.016
	LMGG [102]	1.18	0.73	29.89	0.175
LDC	Vondrick [199]	2.34	0.75	27.96	0.16
	Chung [122]	2.13	0.5	28.22	0.01
	LMGG [102]	1.82	0.57	28.87	0.172
LRW	Vondrick [199]	3.28	0.34	28.03	0.082
	Chung [122]	2.25	0.46	28.06	0.083
	LMGG [102]	1.92	0.53	28.65	0.075
HDTF	MakeItTalk [131]	–	0.802	23.2454	0.1226
	FOMM [45]	–	0.8167	23.4079	0.1345

Table 15 Evaluation of video-based (Part-1) [arrows indicate whether higher (↑) or lower (↓) values are better]

Dataset	Model	CSIM↑	AUCON↑	PRMSE↓	FID↓	AVD↓
VoxCeleb1	FOMM [45]	0.748	0.752	3.66	86	0.044
	OSFV [196]	0.791	0.893	3.01	74	0.028
	DaGAN [60]	0.79	0.88	3.06	87	0.036
	ROME [118]	0.833	0.871	2.64	76	0.016
	FNeVR [200]	0.812	0.884	3.32	82	0.041
	HiDe-NeRF [201]	0.876	0.917	2.62	57	0.012
VoxCeleb2	FOMM [45]	0.68	0.707	4.16	85	0.047
	OSFV [196]	0.711	0.833	3.84	72	0.033
	DaGAN [60]	0.693	0.815	3.93	86	0.04
	ROME [118]	0.71	0.821	3.08	76	0.019
	FNeVR [200]	0.699	0.829	3.9	84	0.047
	HiDe-NeRF [201]	0.787	0.889	2.91	61	0.014
TH-1KH	FOMM [45]	0.723	0.741	3.71	76	0.039
	OSFV [196]	0.787	0.884	3.03	67	0.025
	DaGAN [60]	0.766	0.872	2.98	73	0.035
	ROME [118]	0.781	0.864	2.66	68	0.017
	FNeVR [200]	0.775	0.879	3.39	73	0.037
	HiDe-NeRF [201]	0.828	0.901	2.6	52	0.011

Table 16 Evaluation of video-based (Part-2) [arrows indicate whether higher (↑) or lower (↓) values are better]

DATASET	MODEL	L1↓	PSNR↑	SSIM↑	MS-SSIM↑	FID↓	AKD↓
VoxCeleb2	fs-vid2vid [196]	17.1	20.36	0.71	Nan	85.76	3.41
	FOMM [45]	12.66	23.25	0.77	0.83	73.71	2.14
	Bi-layer [200]	23.95	16.98	0.66	0.66	203.36	5.38
	Neural Talking-Head [119]	10.74	24.37	0.8	0.85	69.13	2.07
TH-1KH	fs-vid2vid [196]	15.18	20.94	0.75	Nan	63.47	11.07
	FOMM [45]	12.3	23.67	0.79	0.83	55.35	3.76
	Bi-layer [200]	12.81	23.13	0.78	Nan	60.58	60.58
	Neural Talking-Head [119]	10.67	24.2	24.2	0.84	52.08	3.74

Object Consistency), PRMSE (Pixel Root Mean Square Error), FID (Fréchet Inception Distance), and AVD (Audio-Visual Distance).

HiDe-NeRF [48] consistently outperforms others in all datasets, achieving the highest scores in CSIM and AUCON and the lowest in PRMSE, FID, and AVD, indicating superior visual quality and audio-visual synchronization. ROME [118] also performs well in AVD, showcasing strong audio-visual synchronization capabilities.

OSFV [196] and DaGAN [60] exhibit competitive performance, with OSFV [196] generally outperforming DaGAN [60]. FOMM [45] ranks lowest among the newer models, whereas FNeVR [160] demonstrates average performance. The TH-1KH [19] dataset is the most challenging, as it typically yields lower performance scores across all models.

The evaluation highlights significant advancements achieved by recent models, particularly HiDe-NeRF [48], in generating high-quality talking head videos with accurate audio-visual synchronization. Notably, the AVD metric reveals substantial differences in performance between the models, indicating a significant area for improvement within the field.

Overall, the tables provide a detailed picture of the current level of THG, emphasizing the importance of evaluating visual quality and motion accuracy, addressing the challenges posed by different datasets, and the continued relevance of AVD and AKD metrics. It also underscores the need for consistent reporting of metrics within the field.

3.4 Evaluation of text-based approaches

A summary of the comparison of various text-based methodologies in various datasets across various evaluation metrics is provided in Tables 17 and 18, which are based on the current literature from the sources [128], which were chosen based on the citation in comparison with others.

The evaluation of text-driven THG models is presented in Table, which includes TFGAN [132] and MMVID [116] across various datasets. The results indicate that MMVID [116] consistently outperforms TFGAN [132] in all datasets, achieving significantly lower Fréchet Video Distance (FVD) and Fréchet Inception Distance (FID) scores, as well as higher CLIP similarity (CLIPSIM) scores. Adding appearance, emotion, and action prompts to CelebV-Text [207] influences performance, with MMVID [208] maintaining a relative advantage over TFGAN [132]. The table also provides a detailed evaluation of various THG models, including MakeItTalk [131], Wav2Lip [70], PC-AVS [130], AVCT, GC-AVT [98], EAMM [202], SadTalker [209], PD-FGC, EAT, StyleTalk [127], and TalkCLIP [125], across MEAD [22], HDTF [24], and VoxCeleb2 [18] datasets. The evaluation focuses on lip synchronization, video quality, and motion accuracy. Models like StyleTalk [127], TalkCLIP [125], and Wav2Lip [70] generally achieve the best SSIM

and Sync(conf) scores, indicating superior visual quality and audio-visual synchronization. The tables highlight the trade-offs between visual quality, lip motion accuracy, and audio-visual synchronization, with models like StyleTalk [127], TalkCLIP [125], and Wav2Lip [70] demonstrating strong performance in balancing these factors. However, there is a large variance in the SSIM scores, showing that visual fidelity is still a large area of research. The tables offer valuable insights into the performance of various THG models, particularly in text-driven scenarios and detailed lip synchronization evaluations. Key takeaways for the review include the significant advancements achieved by models like MMVID [208] in text-driven THG, the importance of evaluating lip synchronization and visual quality using detailed metrics, the influence of dataset characteristics on model performance, the strengths of models like StyleTalk [127], TalkCLIP [125], and Wav2Lip [70] in balancing visual fidelity and audio-visual synchronization, and the continued challenges in achieving perfect lip-sync and high visual fidelity.

3.5 Evaluation of 2D-based approaches

Based on the existing literature, which has been selected based on the citation in comparison with others, we have produced a summary of the comparison of different 2D model-based approaches in different datasets across different assessment metrics in Tables 19 and 20.

Tables 19 and 20 comprehensively evaluate various THG models, including SDA, MakeItTalk [131], Wav2Lip [70], PC-AVS [130], EAMM [202], and Diffused Heads [141], based on the LRW and CREMA datasets. The key metrics assessed include FVD (Fréchet Video Distance), FID (Fréchet Inception Distance), Blinks/s (blinks per second), Blink dur (blink duration), ofM (Optical Flow Magnitude), F-MSE (Facial Mean Squared Error), AV off (Audio-Visual offset), AV Conf. (Audio-Visual Confidence), and WER (Word Error Rate).

DiffusedHeads [141] generally achieve the best FVD scores on both datasets, indicating superior video quality. Wav2Lip [70] excels in FID scores, showcasing outstanding image quality. PC-AVS [130] demonstrates the highest ofM on the LRW dataset, while MakeItTalk [131] and EAMM [202] have lower AV Conf. scores, indicating challenges in audio-visual alignment.

Tables 19 and 20 emphasize the trade-offs between various aspects of THG, such as video quality, lip synchronization, and accuracy in lip-reading. The assortment of metrics illustrates the complexity of evaluating THG. Diffused Heads [141] exhibit a strong capability for producing high-quality videos, while Wav2Lip [70] is noted for generating high-quality images and achieving precise lip synchronization.

Table 17 Evaluation of Text-based (Part-1) [Arrows indicate whether higher (↑) or lower (↓) values are better]

Dataset	Model	FVD ↓	FID ↑	CLIPSIM ↑
MM-Vox	TFGAN [132]	502.28	760.24	0.165
	MMVID [116]	65.79	38.81	0.170
CelebV-HQ	TFGAN [132]	428.04	616.24	0.168
	MMVID [116]	73.65	63.86	0.172
CelebV-Text	TFGAN [132]	403.04	589.24	0.177
	MMVID [116]	66.69	58.70	0.198
CelebV-Text App.+Emo.	TFGAN [132]	442.30	623.17	0.158
	MMVID [116]	82.78	61.58	0.176
CelebV-Text App.+Act.	MMVID-interp [44]	72.87	1.57	1.57
	TFGAN [132]	571.34	784.93	0.154
	MMVID [116]	109.25	82.55	0.174
	MMVID-interp [44]	80.81	70.88	0.176

Table 18 Evaluation of Text-based (Part-2) [Arrows indicate whether higher (↑) or lower (↓) values are better]

Dataset	Model	SSIM ↓	CPBD ↑	F-LMD↓	M-LMD↓	Sync (Conf)↑
MEAD	MakeItTalk [131]	0.73	0.11	3.95	5.39	2.15
	Wav2Lip [202]	0.81	0.16	2.73	3.85	5.41
	PC-AVS [130]	0.51	0.07	5.87	5.03	2.21
	AVCT [203]	0.83	0.14	2.95	5.64	2.56
	GC-AVT [204]	0.34	0.14	8.11	8.43	2.41
	EAMM [202]	0.40	0.08	6.67	6.60	1.42
	SadTalker [198]	0.68	0.16	4.04	4.24	2.87
	PD-FGC [205]	0.51	0.05	5.41	3.94	2.46
	EAT [206]	0.53	0.15	5.63	4.98	2.19
	StyleTalk [127]	0.84	0.16	2.17	3.36	3.51
	TalkCLIP [125]	0.83	0.16	2.42	3.60	3.77
HDTF	MakeItTalk [131]	0.57	0.20	5.12	4.61	3.20
	Wav2Lip [202]	0.59	0.26	5.11	3.84	4.57
	PC-AVS [130]	0.42	0.12	10.7	8.60	4.15
	AVCT [203]	0.74	0.18	3.06	3.83	4.46
	GC-AVT [204]	0.33	0.24	10.7	6.34	4.23
	EAMM [202]	0.37	0.13	7.74	7.67	2.78
	SadTalker [198]	0.73	0.19	6.26	4.18	3.86
	PD-FGC [205]	0.40	0.13	9.99	4.46	4.20
	EAT [206]	0.55	0.18	4.12	4.24	3.95
	StyleTalk [127]	0.80	0.26	2.04	2.50	4.75
	TalkCLIP [125]	0.78	0.25	2.54	2.84	4.69
VoxCeleb2	MakeItTalk [131]	0.52	0.24	6.29	5.15	2.17
	Wav2Lip [202]	0.54	0.30	5.85	4.64	5.70
	PC-AVS [130]	0.36	0.09	12.9	7.42	4.73
	AVCT [203]	0.64	0.23	3.62	3.71	3.89
	EAMM [202]	0.43	0.20	6.36	4.89	2.24
	SadTalker [198]	0.44	0.19	9.12	6.11	4.38
	PD-FGC [205]	0.35	0.12	12.5	8.19	4.64
	EAT [206]	0.47	0.20	5.53	5.88	4.35
	StyleTalk [127]	0.66	0.29	2.92	2.96	4.51
	TalkCLIP [125]	0.67	0.29	2.94	2.99	4.60

Table 19 Evaluation of 2D Model-based (Part-1) [Arrows indicate whether higher (↑) or lower (↓) values are better]

DATASET	MODEL	FVD ↓	FID ↓	Blinks/s	Blink dur	ofM	F-MSE	AV off	AV Conf. ↓	WER ↑
LRW	SDA [199]	198.84	61.95	0.52	0.28	73.82	18.94	1	7.40	0.77
	MakeItTalk [210]	269.29	7.57	0.09	0.28	57.21	3.44	-3	3.16	0.99
	Wav2Lip [33]	366.14	2.83	0.03	0.16	47.12	1.45	-2	6.58	0.51
	PC-AVS [130]	153.12	11.96	0.20	0.16	69.59	17.13	-3	6.24	0.64
	EAMM [202]	172.18	9.28	0.03	0.16	58.46	4.39	-3	3.83	0.95
	Diffused Heads [141]	71.88	3.94	0.35	0.28	70.71	19.69	-2	4.61	0.77
CREMA	SDA [199]	376.48	79.82	0.25	0.26	68.21	6.83	2	5.50	–
	MakeItTalk [210]	256.88	17.26	0.02	0.80	62.36	2.07	-3	3.75	–
	Wav2Lip [33]	193.32	12.57	0.00	–	46.87	1.07	-2	6.68	–
	PC-AVS [130]	333.94	22.53	0.02	0.20	70.36	6.93	-3	6.17	–
	EAMM [202]	6.17	19.40	0.00	–	58.91	1.65	-2	4.26	–
	Diffused Heads [141]	88.61	12.45	0.28	0.36	0.36	6.99	1	4.52	–

Table 20 Evaluation of 2D Model-based (Part-2) [Arrows indicate whether higher (↑) or lower (↓) values are better]

DATASET	MODEL	L1 ↓	PSNR ↑	SSIM ↑	MS-SSIM ↑	FID ↓	AKD ↓
VoxCeleb2	fs-vid2vid [196]	17.10	20.36	0.71	–	85.76	3.41
	FOMM [45]	12.66	23.25	0.77	0.83	73.71	2.14
	Bi-layer [200]	23.95	16.98	0.66	0.66	203.36	5.38
	Neural Talking-Head [119]	10.74	24.37	0.80	0.85	69.13	2.07
TH-1KH	fs-vid2vid [196]	15.18	20.94	0.75	–	63.47	11.07
	FOMM [45]	12.30	23.67	0.79	0.83	55.35	3.76
	FOMM-L [45]	12.81	23.13	0.78	–	60.58	4.04
	Neural Talking-Head [119]	10.67	24.20	0.81	0.84	52.08	3.74

This analysis underscores the importance of meticulously evaluating audio-visual synchronization and reinforces trends identified in previous summaries. It highlights the strengths and weaknesses of different models regarding visual quality, lip synchronization, and lip-reading accuracy. Furthermore, it discusses the challenges of evaluating THG and acknowledges the absence of data in Tables 19 and 20.

3.6 Evaluation of 3D-based approaches

A summary of the comparison of various 3D-based methodologies in various datasets across various evaluation metrics is provided in Tables 21 and 22, which are based on the current literature, which were chosen based on the citation in comparison with others.

Tables 21 and 22 evaluate 3D-aware THG models across multiple datasets, including VoxCeleb [17], CelebV-HQ [25], and TH-1KH [19]. The key metrics assessed include Fréchet Video Distance (FVD), Identity Similarity (ID), Content Distance (CD), and Warping Error (WE).

PV3D [47] consistently performs the best, achieving the lowest scores in FVD and CD and the highest in ID across

all datasets. EG3D+MCG-HD shows strong performance concerning warping error, while both 3DVidGen and 3DVidGen (EG3D) deliver moderate performance. The TH-1KH [19] dataset presents the most challenges, with generally higher FVD and CD scores and lower ID scores. In contrast, the CelebV-HQ [25] dataset yields the best FVD results. It highlights the effectiveness of PV3D [47] in generating high-quality 3D-aware talking head videos.

However, there are significant differences in CD scores, indicating that content consistency remains a substantial area for improvement. Tables 21 and 22 also compare the performance of NeRF-based THG models, including ATVG, Wav2Lip [70], Makeittalk [131], AD-NeRF [88], DFRF, and AE-NeRF [151]. AE-NeRF [151] and DFRF generally achieve the best performance, indicated by the highest PSNR and SSIM scores and the lowest LPIPS scores, showcasing superior image quality and perceptual similarity.

The review emphasizes the advancements in 3D-aware THG, particularly with the PV3D [47] model, and the high-quality results obtained using NeRF-based methods. Additionally, it discusses the challenges of producing top-notch 3D-aware talking head videos and the considerable

Table 21 Evaluation of 3D Model-based (Part-1) [Arrows indicate whether higher (↑) or lower (↓) values are better]

DATASET	MODEL	FVD ↓	ID ↑	CD ↓	WE ↓
VoxCeleb	StyleNeRF+MCG-HD [211]	348.7	0.70	1.08	36.06
	EG3D+MCG-HD [212]	222.1	0.80	1.57	10.57
	3DVidGen [213]	65.5	0.75	3.40	44.55
	3DVidGen (EG3D) [212]	56.3	0.71	3.65	24.55
	PV3D [47]	29.1	0.81	1.34	9.76
CelebV-HQ	StyleNeRF+MCG-HD [211]	134.4	0.80	1.13	38.73
	EG3D+MCG-HD [212]	298.4	0.77	3.34	10.74
	3DVidGen [213]	63.6	0.77	3.80	37.30
	3DVidGen (EG3D) [212]	66.2	0.70	3.83	26.34
	PV3D [47]	39.3	0.81	1.21	8.18
TH-1KH	StyleNeRF+MCG-HD [211]	292.7	0.75	5.34	49.29
	EG3D+MCG-HD [212]	262.4	0.78	1.39	11.54
	3DVidGen [213]	83.0	0.76	4.35	46.47
	3DVidGen (EG3D) [212]	89.8	0.65	4.56	35.48
	PV3D [47]	66.6	0.80	2.33	10.73

Table 22 Evaluation of 3D Model-based Methods (Part-2) [Arrows indicate whether higher (↑) or lower (↓) values are better]

DATASET	MODEL	PSNR ↑	SSIM ↑	LPIPS ↓	LMD ↓
AD-NeRF	ATVG [214]	19.12	0.646	0.523	2.591
	Wav2Lip [70]	29.64	0.843	0.423	2.612
	MakItTalk [131]	22.28	0.655	0.480	10.72
	AD-NeRF [88]	27.73	0.881	0.202	2.603
	DFRF [163]	32.30	0.949	0.080	3.023
	AE-NeRF [151]	32.63	0.949	0.078	2.425

variances in CD and LPIPS scores as potential areas for enhancement.

3.7 Evaluation of NeRF-based approaches

A summary of the comparison of various NeRF-based methodologies in various datasets across various evaluation metrics is provided in Tables 23 and 24, which are based on the current literature, which were chosen based on the citation in comparison with others.

Two comprehensive tables that present an interesting comparison of cutting-edge technologies are used to evaluate NeRF-based THG models. Using the VoxCeleb1 [55] and VoxCeleb2 [18] datasets, Table 7 examines well-known models such as FOMM [45], Bi-Layer, ROME [118], and CVTHead [156]. A range of metrics—such as FID, CSIM, IQA, FPS, PSNR, LPIPS, and MS-SSIM—paints a comprehensive picture of their performance. Leading the pack, CVTHead [156] stands out with its remarkable ability to achieve the lowest FID scores and the highest ratings for IQA, FPS, PSNR, and MS-SSIM. Its exceptional performance in L1 and LPIPS further establishes its dominance in THG by illuminating the developments that enhance motion dynam-

ics and visual fidelity. Simultaneously, FNeVR [160] shows its strength with impressive metrics: it has the highest PSNR and SSIM scores and the lowest FID, L1 loss, AKD, and AED scores. Noting significant differences in AKD scores that can improve motion precision and smoothness, the evaluation emphasizes the significance of examining visual quality and motion accuracy in Face vid2vid [121] and Face vid2vid-S [121] models. The review emphasizes the capacity of FNeVR [160] to produce realistic videos, even as CVTHead [156]’s speed and image quality enhancements are emphasized. The problems of assessing THG are explored in the text, along with the significance of good assessment criteria and the broad range of AKD scores among models. Further research and development in this exciting field are highlighted.

3.8 Evaluation of diffusion-based approaches

Based on the existing literature, which has been selected based on the citation in comparison with others, we have produced a summary of the comparison of different diffusion-based approaches in different datasets across different assessment metrics in Tables 25 and 26.

Table 23 Evaluation of NeRF-Based Methods (Part-1) [Arrows indicate whether higher (↑) or lower (↓) values are better]

DATASET	MODEL	FID ↓	CSIM ↑	IQA ↑	FPS ↑	L1 ↓	PSNR ↑	LPIPS ↓	MS-SSIM ↑
VoxCeleb1	FOMM [45]	39.69	0.592	37.00	64.3	0.048	22.43	0.139	0.836
	Bi-Layer [215]	43.80	0.697	41.40	20.1	0.050	21.48	0.108	0.839
	ROME [118]	29.23	0.717	39.11	12.9	0.048	21.13	0.116	0.838
	CVTHead [156]	25.78	0.675	42.26	24.3	0.041	22.09	0.111	0.840
VoxCeleb2	FOMM [45]	61.28	0.624	36.20	64.3	0.059	20.93	0.165	0.793
	ROME [118]	53.52	0.729	37.34	4.28	0.050	20.75	0.117	0.834
	CVTHead [156]	48.48	0.712	40.27	24.3	0.042	21.37	0.119	0.841

Table 24 Evaluation of NeRF-Based Methods (Part-2) [Arrows indicate whether higher (↑) or lower (↓) values are better]

DATASET	MODEL	FID ↓	L1 ↓	PSNR ↑	LPIPS ↓	SSIM ↑	AKD ↓	AED ↓
VoxCeleb	Bi-layer [215]	219.80	0.1197	15.219	0.4247	0.3968	12.600	0.0546
	FOMM [45]	11.56	0.0450	23.210	0.1099	0.7475	1.383	0.0244
	Face vid2vid [119]	9.142	0.0485	22.642	0.1051	0.7268	1.616	0.0395
	Face vid2vid-S [119]	9.151	0.0445	23.357	0.0901	0.7473	1.421	0.0243
	DaGAN [60]	9.660	0.0462	23.263	0.0981	0.7536	1.441	0.0247
	PIRender [216]	11.88	0.0566	21.040	0.0850	0.6550	2.186	0.2245
	FNeVR [160]	8.443	0.0404	24.292	0.0804	0.7773	1.254	0.0231

Table 25 Evaluation of Diffusion-Based Methods (Part-1) [Arrows indicate whether higher (↑) or lower (↓) values are better]

DATASET	MODEL	FID ↓	CPBD ↑	PSNR ↑	LPIPS ↓	CSIM ↑	LMD ↓	LSE-D ↓
HDTF	Wav2Lip [70]	16.34	0.35	34.81	0.03	0.90	1.43	5.86
	PC-AVS [130]	117.85	0.29	28.23	0.38	0.35	9.36	7.06
	MakeItTalk [131]	66.21	0.43	29.87	0.16	0.82	3.46	10.23
	Audio2Head [100]	65.96	0.37	29.85	0.19	0.71	4.33	7.49
	DiffusedHead [141]	192.74	0.11	28.21	0.274	0.16	22.19	12.37
	DreamTalk [167]	124.51	0.43	29.59	0.20	0.72	2.56	8.37
	MoDiTalker [166]	14.15	0.46	35.82	0.01	0.92	1.38	9.15

Table 26 Evaluation of Diffusion-Based Methods (Part-2) [Arrows indicate whether higher (↑) or lower (↓) values are better]

DATASET	MODEL	FID ↓	FVD ↓	Blinks/s	Blink Dur.	OFMF-MSE	AV Off	AV Conf.	↑ WER ↓
LRW	SDA [199]	61.95	198.84	0.52	0.28	73.82	18.94	1	7.40
	MakeItTalk [131]	7.57	269.29	0.09	0.28	57.21	3.44	-3	3.16
	Wav2Lip [70]	2.83	366.14	0.03	0.16	47.12	1.45	-2	6.58
	PC-AVS [130]	11.96	153.12	0.20	0.16	69.59	17.13	-3	6.24
	EAMM [202]	9.28	172.18	0.03	0.16	58.46	4.39	-3	3.83
	DiffusedHead [141]	3.94	71.88	0.35	0.28	70.71	19.69	-2	4.61
									0.77
CREMA	SDA [199]	79.82	376.48	0.25	0.26	68.21	6.83	2	5.50
	MakeItTalk [131]	17.26	256.88	0.02	0.80	62.36	2.07	-3	3.75
	Wav2Lip [70]	12.57	193.32	0.00	–	46.87	1.07	-2	6.68
	PC-AVS [130]	22.53	333.94	0.02	0.20	70.36	6.93	-3	6.17
	EAMM [202]	19.40	196.82	0.00	–	58.91	1.65	-2	4.26
	DiffusedHead [141]	12.45	88.61	0.28	0.36	64.30	6.99	1	4.52
									–

The table uses the HDTF [24] dataset to assess several diffusion-based THG models. Models like Wav2Lip [70], PC-AVS [130], MakeItTalk [131], Audio2Head [100], DiffusedHead, DreamTalk [167], and MoDiTalker [166] are evaluated in the first table with an emphasis on motion accuracy, lip synchronization, and visual quality. FID (Fréchet Inception Distance), CPBD (Contrast Perception based Blur Detection), PSNR (Peak Signal-to-Noise Ratio), LPIPS (Learned Perceptual Image Patch Similarity), CSIM (Content Similarity), LMD (Lip Motion Distance), and LSE-D (Lip Sync Error Distance) are important metrics. With the lowest FID and LPIPS scores and the highest CPBD, PSNR, and CSIM scores, MoDiTalker [166] performs the best. Wav2Lip [70] also exhibits excellent performance, displaying extremely low LMD and LSE-D values. DiffusedHead displays the worst FID, CSIM, LMD, and LSE-D scores. PC-AVS [130] displays the lowest LPIPS and LMD scores. DreamTalk [167]'s LMD score is low. A thorough analysis of several THG models, such as SDA, MakeItTalk [131], Wav2Lip [70], PC-AVS [130], EAMM [202], and Diffused Heads [141], on the LRW and CREMA datasets is given in the second table. It draws attention to the compromises between THG features, including lip synchronization, video quality, and lip-reading precision. The wide range of metrics demonstrates how difficult it is to assess THG. While Wav2Lip [70] demonstrates a strong ability to produce high-quality images and extremely precise lip-syncing, Diffused Heads [141] demonstrate a strong ability to deliver high-quality video. The review should address the issues in assessing THG, emphasize variances in LMD scores and missing WER data from the CREMA dataset, and highlight the merits and downsides of various models for visual quality, lip synchronization, and lip-reading accuracy.

3.9 Evaluation of 3D animation-based approaches

A summary of the comparison of various 3D animation-based methodologies in various datasets across various evaluation metrics is provided in Table 27, which are based on the current literature, which were chosen based on the citation in comparison with others.

Table 27 contrasts 3D animation-based techniques on three datasets which are TH-1KH [19], CelebV-HQ [25], and VoxCeleb [17]. The top FVD and ID ratings are regularly attained by PV3D [47], demonstrating exceptional visual quality and identification retention. 3DVidGen excels in CD and WE, capturing motion and emotion dynamics, particularly when used with EG3D [212]. In FVD and CD, StyleNeRF+MCG-HD [211] and EG3D+MCG-HD [212] perform inadequately, indicating less expressive or temporally coherent outcomes.

Additionally, 3D animation-based talking head generation has introduced models that blend speech-driven dynamics

with high-fidelity animation, often leveraging rigged avatars, motion capture, and differentiable simulation. Approaches like 3D Gaussian Blendshapes [217] and TalkingGaussian [218] emphasize volumetric and structure-preserving representations using Gaussian primitives to model facial geometry and dynamics in a coherent, persistent 3D space. GaussianTalker [219] extends this by enabling continuous control over expressions with smooth motion transitions. On the expressive and stylized front, AnimateMe [220] and EmoVOCA enhance emotion portrayal in 4D facial sequences, where the latter utilizes speech inputs to render emotional cues in a controllable 3D form. FaceTalk [221] and AVI-Talking [222] adopt diffusion and instruction-guided generation strategies, pushing forward the integration of audio-visual cues for animation. EMOTE [223] employs temporally consistent generative modeling to animate nuanced emotional states. DiffPoseTalk [224] innovates by applying stylistic diffusion to 3D keypoints, enabling fine-grained control over head pose and expression. Imitator [225] introduces personalized 3D modeling, leveraging speaker-specific data for tailoring speech-driven mesh deformation. Finally, FaceXHuBERT [226] employs self-supervised speech embeddings to animate expressive 3D faces without explicit text, focusing on natural expressiveness and identity preservation. Collectively, these models represent a shift toward multimodal, emotionally rich, and controllable 3D talking avatars, with applications spanning virtual humans, real-time dubbing, and affective computing.

3.10 Evaluation of RTR-based approaches

A comprehensive comparison of various RTR-based methodologies across multiple datasets and evaluation metrics is presented in Table 28. The metrics include SSIM, PSNR, and LPIPS for visual quality assessment, FID and E-FID for perceptual realism, LMD and AUE for landmark and expression accuracy, and synchronization measures such as SyncScore, Sync-C, and Sync-D. Models are evaluated primarily on the HDTF dataset [24], providing a standardized benchmark for assessing real-time talking head generation performance.

Table 28 summarizes the performance of RTR-based talking head generation models across several evaluation metrics. Among them, SyncAnimation-One and SyncAnimation-Zero [188] achieve the highest overall performance, reporting superior SSIM (0.7343 and 0.7411), PSNR (21.23 and 21.40), and the lowest LPIPS (0.1543 and 0.1532), indicating high-fidelity and perceptually accurate reconstructions. Hallo2 [93] achieves one of the lowest FID scores (16.616), demonstrating strong realism and identity preservation, while maintaining competitive synchronization metrics (Sync-C = 7.379). SadTalker [209] exhibits robust temporal consistency, with a relatively low FVD (293.084), outperforming models like EchoMimic [108], which shows

Table 27 Evaluation of 3D Animation-Based [Arrows indicate whether higher (↑) or lower (↓) values are better]

DATASET	MODEL	FVD ↓	ID ↑	CD ↑	WE ↑
VoxCeleb	StyleNeRF+MCG-HD [211]	348.7	0.70	1.08	36.06
	EG3D+MCG-HD [212]	222.1	0.80	1.57	10.57
	3DVidGen [213]	65.5	0.75	3.40	44.55
	3DVidGen (EG3D) [213]	56.3	0.71	3.65	24.55
	PV3D [47]	29.1	0.81	1.34	9.76
CelebV-HQ	StyleNeRF+MCG-HD [211]	134.4	0.80	1.13	38.73
	EG3D+MCG-HD [212]	298.4	0.77	3.34	10.74
	3DVidGen [213]	63.6	0.77	3.80	37.30
	3DVidGen (EG3D) [213]	66.2	0.70	3.83	26.34
	PV3D [47]	39.3	0.81	1.21	8.18
TH-1KH	StyleNeRF+MCG-HD [211]	292.7	0.75	5.34	49.29
	EG3D+MCG-HD [212]	262.4	0.78	1.39	11.54
	3DVidGen [213]	83.0	0.76	4.35	46.47
	3DVidGen (EG3D) [213]	89.8	0.65	4.56	35.48
	PV3D [47]	66.6	0.80	2.33	10.73

the highest FVD (532.733) and LPIPS (0.4524), indicating limitations in temporal smoothness and photorealism. Hallo [93] provides balanced performance with SSIM of 0.6305 and a reasonable FID of 35.83 but underperforms compared to more recent approaches on synchronization metrics. In contrast, V-Express [86] focuses on expressive portrait generation but achieves lower PSNR (17.39) and higher FID (46.69), showing a trade-off between stylization and realism. Wav2Lip [70], while being one of the earliest audio-visual synchronization models and achieving a strong SyncScore of 9.2332, underperforms in perceptual quality compared to newer RTR-based frameworks, highlighting the improvements made by recent architectures. In summary, RTR-based approaches demonstrate significant progress in real-time talking head generation, delivering substantial improvements in visual quality, perceptual realism, and temporal consistency. Architectures based on U-Net and transformer-driven diffusion models, such as SyncAnimation, achieve strong results across multiple evaluation metrics. Additionally, methods employing Gaussian-based and hybrid rendering techniques contribute to improved identity preservation and synchronization. Future research is expected to focus on unified optimization strategies that integrate multimodal conditioning to enhance expressiveness, generalization, and interactive performance.

4 Challenges and future directions

The field of THG is evolving, presenting both challenges and opportunities for future research. A significant issue is the dependence on large, pre-trained models such as GANs and transformers, which can introduce biases from unbalanced

datasets and limit adaptability in resource-constrained environments. There is an urgent need for modular architectures that facilitate independent training of system components, thereby reducing overfitting and enhancing flexibility. Handling pose diversity and occlusions poses another challenge, as current systems struggle with extreme head rotations and dynamic occlusions, largely due to biased datasets that focus on frontal imagery. It is essential to improve the quality and diversity of training data, particularly through multi-view and high-frame-rate recordings. Additionally, advancements in disentangling identity, expression, and pose can help generalize results to rare conditions. The lack of multilingual capabilities is another significant concern; most models are trained on English-centric datasets, resulting in poor performance in other languages. Expanding access to high-quality multilingual datasets and employing self-supervised learning can enhance cross-lingual performance. Maintaining temporal consistency and enabling real-time generation also present challenges, as many systems display flickering or drift in longer sequences. Innovative architectures, such as temporal transformers and LSTM networks, along with techniques aimed at optimizing model efficiency, can improve real-time deployment. The future of THG may reside in hybrid architectures that integrate 2D keypoint and 3D modeling techniques, allowing for customized applications. Finally, refining evaluation metrics to capture the holistic quality of THG outputs—beyond traditional visual metrics—is crucial for accurately assessing system effectiveness.

Table 28 Evaluation of RTR-Based [Arrows indicate whether higher (↑) or lower (↓) values are better]

DATASET	MODEL	SSIM↑	PSNR↑	LPIPS↓	FID↓	LMD↓	AUE↓	SyncScore↑	FVD↓	Sync-C↑	Sync-D↓	E-FID↓
HDTF	Wav2Lip [70]	0.6571	18.8071	0.2523	40.3604	5.2161	4.5499	9.2332	—	—	—	—
	SadTalker [209]	—	—	—	21.924	—	—	—	293.084	7.399	7.812	6.881
	EchoMimic [108]	0.4562	14.4399	0.4524	59.3261	8.0643	5.1434	6.0574	532.733	5.930	9.143	11.051
	Hallo [93]	0.6305	17.9117	0.2678	35.8346	6.0491	4.3952	7.2458	366.066	7.268	7.714	7.081
	Hallo2 [93]	—	—	—	16.616	—	—	—	239.517	7.379	7.697	6.702
	V-Express [86]	0.6121	17.3918	0.2842	46.6975	6.9958	5.4740	7.4739	—	—	—	—
	SyncAnimation-One [188]	0.7343	21.2323	0.1543	20.0567	3.2215	3.5485	7.1389	—	—	—	—
	SyncAnimation-Zero [188]	0.7411	21.4006	0.1532	21.7353	3.1878	3.5515	7.1499	—	—	—	—

5 Conclusion

THG has significantly advanced over the past decade, transforming from basic rule-based frameworks to sophisticated deep learning models. Early 2D-based approaches, such as keypoint-driven synthesis and convolutional neural networks, faced challenges like depth ambiguity, poor generalization, and insufficient preservation of facial details. The transition to 3D-based models and volumetric representations improved rendering realism and viewpoint control, allowing for free-viewpoint generation and more accurate facial geometry and head motion modeling. Diffusion models have emerged as a powerful generative framework but are computationally expensive and limited for real-time or low-resource environments. 3D animation frameworks have gained popularity in domains requiring precise control over facial movements. Evaluation of THG systems has evolved to incorporate synchronization-focused metrics, Fréchet Video Distance, and semantic similarity tools. Benchmarking THG models across publicly available datasets has established common evaluation standards but still faces limitations in speaker diversity, resolution quality, expressive richness, and multilingual representation. Talking head generation (THG) models face challenges in addressing demographic biases, robustness under extreme conditions, and maintaining temporal consistency. These models rely on English-centric datasets, limiting their applicability in linguistically diverse contexts. Preserving temporal consistency is also challenging, especially in long-form content generation or real-time streaming applications. Real-time THG is still an open challenge, especially for high-resolution video synthesis on consumer-grade hardware. Ethical concerns surrounding THG misuse have intensified, with deepfakes, misinformation, identity theft, and consent violations. The absence of standardized regulatory frameworks governing synthetic media's creation, distribution, and attribution exacerbates these issues. Developing watermarking systems, authenticity verification tools, and ethical guidelines is critical for the responsible use of THG. Public awareness and education about the capabilities and limitations of THG are needed, as well as broader interdisciplinary collaborations. Sustainable progress in THG will depend on addressing technical bottlenecks, expanding dataset diversity, improving computational efficiency, and fostering ethical innovation.

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TS contributed through provision of resources, supporting the project's execution. All authors reviewed and approved the final version of the manuscript and agreed to be accountable for all aspects of the work.

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Data Availability The datasets and related resources used in this study are openly available on GitHub and will be maintained and updated regularly. Interested researchers can access the data at: <https://github.com/VineetKumarRakesh/thg>

Declarations

Conflict of interest The authors declare no conflict of interest.

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